

Competitive Programming Notebook

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1 Miscellaneous

1.1 Template

```
1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 //define int long long
6 #define ll long long
7 #define ld long double
8 #define fi first
9 #define se second
10 #define pb push_back
11 #define eb emplace_back
12 #define all(x) x.begin(), x.end()
13 #define rall(x) x.rbegin(), x.rend()
14 #define ones(n) __builtin_popcountll(n)
15 #define msb(n) (63 - __builtin_clzll(n))
16 #define lsb(n) __builtin_ctzll(n)
17
18 const int N = 2e5 + 5, M = 1e3 + 5, LOG = 31;
19 const int inf = 0x3f3f3f3f;
20 const ll llinf = 0x3f3f3f3f3f3f3f3f;
21 const int MOD = 1e9 + 7;
22 const double eps = 1e-9, PI = acos(-1);
23
24 void testCase() {
25 }
26
27 void preCompute() {
28 }
29
30 int32_t main() {
31 #ifdef C_Lion
32     freopen("input.txt", "r", stdin);
33     freopen("output.txt", "w", stdout);
34     freopen("errors.txt", "w", stderr);
35 #else
36     //     freopen("input.in", "r", stdin);
37     //     freopen("output.out", "w", stdout);
38 #endif
39
40     ios_base::sync_with_stdio(false), cin.tie(nullptr), cout.tie(
        nullptr);
41     cout << fixed << setprecision(static_cast<int32_t>(-log10(eps)));
42     preCompute();
43
44     int tc = 1;
45     cin >> tc;
46     for (int TC = 1; TC <= tc; TC++) {
47         // cout << "Case #" << TC << ": ";
48         testCase();
49     }
50 }
51
52 /*
53
54
55
56
57
58 */
```

1.2 fastIO

```
1 const int MOD = 1e9 + 7;
2 const int BUF_SZ = 1 << 15;
3
4 inline namespace Input {
5     char buf[BUF_SZ];
6     int pos;
7     int len;
8     char next_char() {
9         if (pos == len) {
10             pos = 0;
11             len = (int)fread(buf, 1, BUF_SZ, stdin);
12             if (!len) { return EOF; }
13         }
14         return buf[pos++];
15     }
16
17     int read_int() {
18         int x;
19         char ch;
20         int sgn = 1;
21         while (!isdigit(ch = next_char())) {
22             if (ch == '-') { sgn *= -1; }
23         }
24         x = ch - '0';
25         while (isdigit(ch = next_char())) { x = x * 10 + (ch - '0'); }
26         return x * sgn;
27     }
28 } // namespace Input
29 inline namespace Output {
30     char buf[BUF_SZ];
31     int pos;
32
33     void flush_out() {
```

```
34         fwrite(buf, 1, pos, stdout);
35         pos = 0;
36     }
37
38     void write_char(char c) {
39         if (pos == BUF_SZ) { flush_out(); }
40         buf[pos++] = c;
41     }
42
43     void write_int(int x) {
44         static char num_buf[100];
45         if (x < 0) {
46             write_char('-');
47             x *= -1;
48         }
49         int len = 0;
50         for (; x >= 10; x /= 10) { num_buf[len++] = (char)('0' + (x %
            10)); }
51         write_char((char)('0' + x));
52         while (len) { write_char(num_buf[--len]); }
53         write_char('\n');
54     }
55
56     // auto-flush output when program exits
57     void init_output() { assert(atexit(flush_out) == 0); }
58 } // namespace Output
```

1.3 grid

```
1 int dx[] = {0, 0, -1, 1, -1, -1, 1, 1};
2 int dy[] = {-1, 1, 0, 0, -1, 1, -1, 1};
3 char di[] = {'L', 'R', 'U', 'D'};
```

1.4 random

```
1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 template<typename T>
3 T random(T l, T r) {
4     return uniform_int_distribution(l, r)(rng);
5 }
6
7 vector<int> random(int n) {
8     vector<int> v(n);
9     // shuffle 1
10    shuffle(v.begin(), v.end(), rng);
11    // shuffle 2
12    for (int i = 1; i < n; i++)
13        swap(v[i], v[uniform_int_distribution<int>(0, i)(rng)]);
14    return v;
15 }
```

1.5 coordinateCompression

```
1 // Coordinate Compression - O(N log N)
2 template<typename T>
3 struct Compress {
4     vector<T> v;
5
6     Compress(vector<T>& a) : v(a) {
7         sort(all(v));
8         v.erase(unique(all(v)), v.end());
9         for (auto &x : a) {
10             x = lower_bound(all(v), x) - v.begin();
11         }
12     }
13
14     T operator[](int i) const {
15         return v[i];
16     }
17 };
```

1.6 grayCode

```
1 // Gray Code Generator - O(2^N)
2 // g(n) = n ^ (n >> 1)
3 vector<int> grayCode(int n) {
4     vector<int> res(1 << n);
5     for (int i = 0; i < (1 << n); i++) {
6         res[i] = i ^ (i >> 1);
7     }
8     return res;
9 }
```

1.7 customHashHashTable

```
1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3
4 // Custom Hash to prevent anti-hash tests in unordered_map /
5 // gp_hash_table
6 struct custom_hash {
7     static uint64_t splitmix64(uint64_t x) {
8         x += 0x9e3779b97f4a7c15;
9         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
10        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
11        return x ^ (x >> 31);
12    }
13
14    size_t operator()(uint64_t x) const {
15        static const uint64_t FIXED_RANDOM = chrono::steady_clock::now
            ().time_since_epoch().count();
16        return splitmix64(x + FIXED_RANDOM);
17    };
18
19    // Usage: gp_hash_table<ll, int, custom_hash> table;
20    // unordered_map<ll, int, custom_hash> safe_map;
```

1.8 pragmas

```

1 // GCC Optimizations
2 #pragma GCC optimize("O3,unroll-loops")
3 #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")

```

Data Structures (DS)

2.1 Segment Tree

2.1.1 SegmentTree

```

1 struct SEG {
2     ll sum = 0;
3     SEG() {
4     }
5     SEG(ll x){
6         sum = x;
7     }
8 };
9 struct segTree {
10     vector<SEG> seg;
11     int sz = 1,n;
12     segTree(int nn){
13         n = nn;
14         while (sz < nn)
15             sz *= 2;
16         seg.assign(2 * sz , SEG());
17     }
18
19     SEG merge(SEG seg1, SEG seg2) {
20         SEG ret;
21         ret.sum = seg1.sum + seg2.sum;
22         return ret;
23     }
24
25     void build(vector<int> &v,int x, int lx, int rx) {
26         if (lx == rx) {
27             seg[x] = SEG(v[lx]);
28             return;
29         }
30         int mid = (lx + rx) / 2;
31         int lf = 2 * x + 1, rt = 2 * x + 2;
32
33         build(v,lf, lx, mid);
34         build(v,rt, mid + 1, rx);
35         seg[x] = merge(seg[lf], seg[rt]);
36     }
37     void build(vector<int> &v){
38         build(v,0, 0, n-1);
39     }
40
41     void update(int i, ll val, int x, int lx, int rx) {
42         if (lx == rx) {
43             seg[x] = SEG(val);
44             return;
45         }
46         int mid = (lx + rx) / 2,lf = 2*x+1,rt= 2*x+2;
47         if(i <= mid)
48             update(i, val, lf, lx, mid);
49         else
50             update(i, val, rt, mid + 1, rx);
51         seg[x] = merge(seg[lf] , seg[rt]);
52     }
53
54     void update(int i, ll val) {
55         update(i, val, 0, 0, n-1);
56     }
57
58     SEG query(int l, int r, int x, int lx, int rx) {
59         if (l <= lx && r >= rx)
60             return seg[x];
61         if (l > rx || r < lx)
62             return SEG();
63
64         int mid = (lx + rx) / 2,lf = 2 * x + 1,rt = 2*x+2;
65
66         return merge(query(l, r, lf, lx, mid) , query(l, r, rt, mid +
67             1, rx));
68     }
69
70     SEG query(int l, int r) {
71         return query(l, r, 0, 0, n-1);
72     }
73 };

```

2.1.2 LazySegmentTree

```

1 struct SEG {
2     ll sum = 0, lz = 0;
3
4     SEG() {
5     }
6
7     SEG(ll x) {
8         sum = x;
9     }
10 };
11
12 struct segTree {
13     vector<SEG> seg;
14     int sz = 1, n;
15     ll NO_OP = 0;
16
17     segTree(int nn) {
18         n = nn;
19         while (sz < nn)
20             sz *= 2;
21         seg.assign(2 * sz, SEG());
22     }
23
24     SEG merge(SEG seg1, SEG seg2) {
25         SEG ret;
26         ret.sum = seg1.sum + seg2.sum;
27         return ret;
28     }
29
30     void build(vector<int> &v,int x, int lx, int rx) {
31         if (lx == rx) {
32             seg[x] = SEG(v[lx]);
33             return;
34         }
35         int mid = (lx + rx) / 2;
36         int lf = 2 * x + 1, rt = 2 * x + 2;
37
38         build(v,lf, lx, mid);
39         build(v,rt, mid + 1, rx);
40         seg[x] = merge(seg[lf], seg[rt]);
41     }
42
43     void build(vector<int> &v){
44         build(v,0, 0, n-1);
45     }
46
47     void update(int i, ll val, int x, int lx, int rx) {
48         if (lx == rx) {
49             seg[x] = SEG(val);
50             return;
51         }
52         int mid = (lx + rx) / 2,lf = 2*x+1,rt= 2*x+2;
53         if(i <= mid)
54             update(i, val, lf, lx, mid);
55         else
56             update(i, val, rt, mid + 1, rx);
57         seg[x] = merge(seg[lf] , seg[rt]);
58     }
59
60     void update(int i, ll val) {
61         update(i, val, 0, 0, n-1);
62     }
63
64     SEG query(int l, int r, int x, int lx, int rx) {
65         if (l <= lx && r >= rx)
66             return seg[x];
67         if (l > rx || r < lx)
68             return SEG();
69
70         int mid = (lx + rx) / 2,lf = 2 * x + 1,rt = 2*x+2;
71
72         return merge(query(l, r, lf, lx, mid) , query(l, r, rt, mid +
73             1, rx));
74     }
75
76     SEG query(int l, int r) {
77         return query(l, r, 0, 0, n-1);
78     }
79 };

```

```

28 }
29
30 void build(vector<int> &v, int x, int lx, int rx) {
31     if (lx == rx) {
32         seg[x] = SEG(v[lx]);
33         return;
34     }
35     int mid = (lx + rx) / 2;
36     int lf = 2 * x + 1, rt = 2 * x + 2;
37
38     build(v, lf, lx, mid);
39     build(v, rt, mid + 1, rx);
40     seg[x] = merge(seg[lf], seg[rt]);
41 }
42
43 void build(vector<int> &v) {
44     build(v, 0, 0, n - 1);
45 }
46
47 void push(int x, int lx, int rx) {
48     if (seg[x].lz != NO_OP) {
49         seg[x].sum += seg[x].lz * (rx - lx + 1);
50         int lf = 2 * x + 1, rt = 2 * x + 2;
51         if (lx != rx) {
52             seg[lf].lz += seg[x].lz;
53             seg[rt].lz += seg[x].lz;
54         }
55         seg[x].lz = NO_OP;
56     }
57 }
58
59 void update(int l, int r, ll val, int x, int lx, int rx) {
60     push(x, lx, rx);
61     if (l > rx || r < lx)
62         return;
63     if (l <= lx && r >= rx) {
64         seg[x].lz += val;
65         push(x, lx, rx);
66         return;
67     }
68     int mid = (lx + rx) / 2, lf = 2 * x + 1, rt = 2 * x + 2;
69     update(l, r, val, lf, lx, mid);
70     update(l, r, val, rt, mid + 1, rx);
71     seg[x] = merge(seg[lf], seg[rt]);
72 }
73
74 void update(int l, int r, ll val) {
75     update(l, r, val, 0, 0, sz - 1);
76 }
77
78 SEG query(int l, int r, int x, int lx, int rx) {
79     push(x, lx, rx);
80     if (l <= lx && r >= rx)
81         return seg[x];
82     if (l > rx || r < lx)
83         return SEG();
84
85     int mid = (lx + rx) / 2, lf = 2 * x + 1, rt = 2 * x + 2;
86
87     return merge(query(l, r, lf, lx, mid), query(l, r, rt, mid + 1,
88         rx));
89 }
90
91 SEG query(int l, int r) {
92     return query(l, r, 0, 0, sz - 1);
93 }
94 };

```

2.1.3 PersistentSegmentTree

```

1 struct Node {
2     Node *l, *r;
3     ll val = 0;
4
5     Node(int val) : l(NULL), r(NULL), val(val) {
6     }
7
8     Node() : l(NULL), r(NULL) {
9     }
10
11     Node(Node* l, Node* r) : l(l), r(r) {
12         if (l != NULL) val += l->val;
13         if (r != NULL) val += r->val;
14     }
15
16     void addChild() {
17         l = new Node();
18         r = new Node();
19     }
20 };
21
22 struct PST {
23     int n;
24
25     PST(int n) : n(n + 1) {
26     }
27
28     Node merge(Node x, Node y) {
29         Node ret;
30         ret.val = x.val + y.val;
31         return ret;
32     }
33
34     Node* update(Node* v, int i, int val, int lx, int rx) {
35         if (lx == rx) return new Node(v->val + val);
36         int mid = (lx + rx) / 2;
37         if (!v->l) v->addChild();
38         if (i <= mid) return new Node(update(v->l, i, val, lx, mid), v
39             ->r);
40         return new Node(v->l, update(v->r, i, val, mid + 1, rx));
41     }
42
43     Node* update(Node* v, int i, int val) { return update(v, i, val, 0,
44         n - 1); }
45 };

```

```

43 Node query(Node* v, int l, int r, int lx, int rx) {
44     if (l > rx || r < lx) return {};
45     if (l <= lx && r >= rx) return *v;
46     if (!v->l) v->addChild();
47     int mid = (lx + rx) / 2;
48     return merge(query(v->l, l, r, lx, mid), query(v->r, l, r, mid
49         + 1, rx));
50 }
51
52 Node query(Node* v, int l, int r) { return query(v, l, r, 0, n - 1)
53     ; }
54
55 int getKth(Node* a, Node* b, int k, int lx, int rx) {
56     if (lx == rx) return lx;
57     if (!a->l) a->addChild();
58     if (!b->l) b->addChild();
59     int rem = b->l->val - a->l->val;
60     int mid = (lx + rx) / 2;
61     if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
62     return getKth(a->r, b->r, k - rem, mid + 1, rx);
63 }
64
65 int getKth(Node* a, Node* b, int k) { return getKth(a, b, k, 0, n -
66     1); }
67
68 };

```

2.1.4 PersistentSegTreeVector

```

1 struct Node {
2     Node *l, *r;
3     int val = 0;
4
5     Node(int val): l(NULL), r(NULL), val(val) {}
6     Node(): l(NULL), r(NULL) {}
7     Node(Node *l, Node *r): l(l), r(r) {
8         if (l != NULL) val += l->val;
9         if (r != NULL) val += r->val;
10    }
11
12    void addChild() {
13        l = new Node();
14        r = new Node();
15    }
16 };
17
18 struct PST {
19     int n;
20     PST(int n): n(n + 1) {}
21
22     Node merge(Node x, Node y) {
23         Node ret;
24         ret.val = x.val + y.val;
25         return ret;
26    }
27
28     Node *set(Node *v, int i, int val, int lx, int rx) {
29         if (lx == rx) return new Node(val);
30         int mid = (lx + rx) / 2;
31         if (!v->l) v->addChild();
32         if (i <= mid) return new Node(set(v->l, i, val, lx, mid), v->r)
33             ;
34         return new Node(v->l, set(v->r, i, val, mid + 1, rx));
35    }
36
37     Node *set(Node *v, int i, int val) { return set(v, i, val, 0, n -
38         1); }
39
40     // [l, r] r is included
41     Node query(Node *v, int l, int r, int lx, int rx) {
42         if (l > rx || r < lx) return {};
43         if (l <= lx && r >= rx) return *v;
44         if (!v->l) v->addChild();
45         int mid = (lx + rx) / 2;
46         return merge(query(v->l, l, r, lx, mid), query(v->r, l, r, mid
47             + 1, rx));
48    }
49
50     Node query(Node *v, int l, int r) { return query(v, l, r, 0, n - 1)
51         ; }
52
53     int getKth(Node *a, Node *b, int k, int lx, int rx) {
54         if (lx == rx) return lx;
55         if (!a->l) a->addChild();
56         if (!b->l) b->addChild();
57         int rem = b->l->val - a->l->val;
58         int mid = (lx + rx) / 2;
59         if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
60         return getKth(a->r, b->r, k - rem, mid + 1, rx);
61    }
62
63     int getKth(Node *a, Node *b, int k) { return getKth(a, b, k, 0, n -
64         1); }
65
66 };

```

2.1.5 SegmentTreeBeats

```

1 struct Node {
2     int mn;
3     int mnCnt;
4     int secondMn;
5
6     int mx;
7     int mxCnt;
8     int secondMx;
9
10    int sum;
11
12    int lazyAdd;
13    int lazyEqual;
14    bool isLazyEqual;
15
16    Node() {
17        mx = -inf;

```

```

18        secondMx = -inf;
19        mxCnt = 0;
20
21        mn = inf;
22        secondMn = inf;
23        mnCnt = 0;
24
25        sum = 0;
26
27        lazyAdd = 0;
28        lazyEqual = 0;
29        isLazyEqual = 0;
30    };
31
32    Node(int x) {
33        mx = x;
34        secondMx = -inf;
35        mxCnt = 1;
36
37        mn = x;
38        secondMn = inf;
39        mnCnt = 1;
40
41        sum = x;
42
43        lazyAdd = 0;
44        lazyEqual = 0;
45        isLazyEqual = 0;
46    }
47 };
48
49 struct SegTree {
50     int tree_size;
51     vector<Node> seg_data;
52
53     SegTree(int n) {
54         tree_size = 1;
55         while (tree_size < n) tree_size *= 2;
56         seg_data.resize(2 * tree_size, Node());
57     }
58
59     Node merge(Node& lf, Node& ri) {
60         Node ans = Node();
61
62         ans.sum = lf.sum + ri.sum;
63
64         ans.mx = max(lf.mx, ri.mx);
65         ans.secondMx = max(lf.secondMx, ri.secondMx);
66         ans.mxCnt = 0;
67         if (lf.mx == ans.mx) ans.mxCnt += lf.mxCnt;
68         else ans.secondMx = max(ans.secondMx, lf.mx);
69         if (ri.mx == ans.mx) ans.mxCnt += ri.mxCnt;
70         else ans.secondMx = max(ans.secondMx, ri.mx);
71
72         ans.mn = min(lf.mn, ri.mn);
73         ans.secondMn = min(lf.secondMn, ri.secondMn);
74         ans.mnCnt = 0;
75         if (lf.mn == ans.mn) ans.mnCnt += lf.mnCnt;
76         else ans.secondMn = min(ans.secondMn, lf.mn);
77         if (ri.mn == ans.mn) ans.mnCnt += ri.mnCnt;
78         else ans.secondMn = min(ans.secondMn, ri.mn);
79
80         return ans;
81     }
82
83     void init(vector<int>& nums, int ni, int lx, int rx) {
84         if (rx - lx == 1) {
85             if (lx < nums.size())
86                 seg_data[ni] = Node(nums[lx]);
87             return;
88         }
89
90         int mid = lx + (rx - lx) / 2;
91         init(nums, 2 * ni + 1, lx, mid);
92         init(nums, 2 * ni + 2, mid, rx);
93
94         seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
95             2]);
96     }
97
98     void init(vector<int>& nums) {
99         init(nums, 0, 0, tree_size);
100    }
101
102     void doPushEq(int ni, int lx, int rx, int x) {
103         seg_data[ni].mn = seg_data[ni].mx = seg_data[ni].lazyEqual = x;
104         seg_data[ni].isLazyEqual = 1;
105         seg_data[ni].mnCnt = seg_data[ni].mxCnt = rx - lx;
106         seg_data[ni].secondMx = -inf;
107         seg_data[ni].secondMn = inf;
108
109         seg_data[ni].sum = x * (rx - lx);
110         seg_data[ni].lazyAdd = 0;
111     }
112
113     void doPushMinEq(int ni, int lx, int rx, int x) {
114         if (seg_data[ni].mn >= x) doPushEq(ni, lx, rx, x);
115         else if (seg_data[ni].mx > x) {
116             if (seg_data[ni].secondMn == seg_data[ni].mx)
117                 seg_data[ni].secondMn = x;
118             seg_data[ni].sum -= (seg_data[ni].mx - x) * seg_data[ni].
119                 mxCnt;
120             seg_data[ni].mx = x;
121         }
122     }
123
124     void doPushMaxEq(int ni, int lx, int rx, int x) {
125         if (seg_data[ni].mx <= x) doPushEq(ni, lx, rx, x);
126         else if (seg_data[ni].mn < x) {
127             if (seg_data[ni].secondMx == seg_data[ni].mn)
128                 seg_data[ni].secondMx = x;
129
130             seg_data[ni].sum += (x - seg_data[ni].mn) * seg_data[ni].
131                 mnCnt;

```

```

129     seg_data[ni].mn = x;
130 }
131 }
132
133 void doPushSum(int ni, int lx, int rx, int x) {
134     if (seg_data[ni].mn == seg_data[ni].mx)
135         doPushEq(ni, lx, rx, seg_data[ni].mn + x);
136     else {
137         seg_data[ni].mx += x;
138         if (seg_data[ni].secondMx != -inf)
139             seg_data[ni].secondMx += x;
140
141         seg_data[ni].mn += x;
142         if (seg_data[ni].secondMn != inf)
143             seg_data[ni].secondMn += x;
144
145         seg_data[ni].sum += (rx - lx) * x;
146         seg_data[ni].lazyAdd += x;
147     }
148 }
149
150 void propagete(int ni, int lx, int rx) {
151     if (rx - lx == 1) return;
152
153     int mid = lx + (rx - lx) / 2;
154     if (seg_data[ni].isLazyEqual) {
155         doPushEq(2 * ni + 1, lx, mid, seg_data[ni].lazyEqual);
156         doPushEq(2 * ni + 2, mid, rx, seg_data[ni].lazyEqual);
157         seg_data[ni].lazyEqual = seg_data[ni].isLazyEqual = 0;
158     }
159     else {
160         doPushSum(2 * ni + 1, lx, mid, seg_data[ni].lazyAdd);
161         doPushSum(2 * ni + 2, mid, rx, seg_data[ni].lazyAdd);
162         seg_data[ni].lazyAdd = 0;
163
164         doPushMinEq(2 * ni + 1, lx, mid, seg_data[ni].mx);
165         doPushMinEq(2 * ni + 2, mid, rx, seg_data[ni].mx);
166
167         doPushMaxEq(2 * ni + 1, lx, mid, seg_data[ni].mn);
168         doPushMaxEq(2 * ni + 2, mid, rx, seg_data[ni].mn);
169     }
170 }
171
172 void minimize(int l, int r, int x, int ni, int lx, int rx) {
173     if (rx <= 1 || lx >= r || seg_data[ni].mx <= x)
174         return;
175     if (lx >= 1 && rx <= r && seg_data[ni].secondMx < x) {
176         doPushMinEq(ni, lx, rx, x);
177         return;
178     }
179
180     propagete(ni, lx, rx);
181
182     int mid = lx + (rx - lx) / 2;
183     minimize(l, r, x, 2 * ni + 1, lx, mid);
184     minimize(l, r, x, 2 * ni + 2, mid, rx);
185
186     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni + 2]);
187 }
188
189 void minimize(int l, int r, int x) {
190     minimize(l, r, x, 0, 0, tree_size);
191 }
192
193 void maximize(int l, int r, int x, int ni, int lx, int rx) {
194     if (rx <= 1 || lx >= r || seg_data[ni].mn >= x)
195         return;
196     if (lx >= 1 && rx <= r && seg_data[ni].secondMn > x) {
197         doPushMaxEq(ni, lx, rx, x);
198         return;
199     }
200
201     propagete(ni, lx, rx);
202
203     int mid = lx + (rx - lx) / 2;
204     maximize(l, r, x, 2 * ni + 1, lx, mid);
205     maximize(l, r, x, 2 * ni + 2, mid, rx);
206
207     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni + 2]);
208 }
209
210 void maximize(int l, int r, int x) {
211     maximize(l, r, x, 0, 0, tree_size);
212 }
213
214 void add(int l, int r, int x, int ni, int lx, int rx) {
215     if (rx <= 1 || lx >= r)
216         return;
217     if (lx >= 1 && rx <= r) {
218         doPushSum(ni, lx, rx, x);
219         return;
220     }
221
222     propagete(ni, lx, rx);
223
224     int mid = lx + (rx - lx) / 2;
225     add(l, r, x, 2 * ni + 1, lx, mid);
226     add(l, r, x, 2 * ni + 2, mid, rx);
227
228     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni + 2]);
229 }
230
231 void add(int l, int r, int x) {
232     add(l, r, x, 0, 0, tree_size);
233 }
234
235 void set(int l, int r, int x, int ni, int lx, int rx) {
236     if (rx <= 1 || lx >= r)
237         return;
238     if (lx >= 1 && rx <= r) {
239         doPushEq(ni, lx, rx, x);

```

```

240         return;
241     }
242
243     propagete(ni, lx, rx);
244
245     int mid = lx + (rx - lx) / 2;
246     set(l, r, x, 2 * ni + 1, lx, mid);
247     set(l, r, x, 2 * ni + 2, mid, rx);
248
249     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni + 2]);
250 }
251
252 void set(int l, int r, int x) {
253     set(l, r, x, 0, 0, tree_size);
254 }
255
256 Node get_range(int l, int r, int ni, int lx, int rx) {
257     propagete(ni, lx, rx);
258
259     if (lx >= 1 && rx <= r)
260         return seg_data[ni];
261     if (rx <= 1 || lx >= r)
262         return Node();
263
264     int mid = (lx + rx) / 2;
265
266     Node lf = get_range(l, r, 2 * ni + 1, lx, mid);
267     Node ri = get_range(l, r, 2 * ni + 2, mid, rx);
268     return merge(lf, ri);
269 }
270
271 int get_sum(int l, int r) {
272     return get_range(l, r, 0, 0, tree_size).sum;
273 }
274
275 int get_mx(int l, int r) {
276     return get_range(l, r, 0, 0, tree_size).mx;
277 }
278
279 int get_mn(int l, int r) {
280     return get_range(l, r, 0, 0, tree_size).mn;
281 }
282 };

```

2.2 Binary Indexed Tree (BIT)

2.2.1 FenwickTree

```

1 // Fenwick Tree (Binary Indexed Tree) - 0-based indexing
2 // Space: O(N), Time: O(log N) per update/query
3 template<typename T = ll>
4 struct FenwickTree {
5     int n;
6     vector<T> tree;
7
8     FenwickTree(int n = 0) : n(n), tree(n + 1, 0) {}
9
10    void add(int i, T delta) {
11        for (i++; i <= n; i += i & -i) tree[i] += delta;
12    }
13
14    void set(int i, T val) {
15        add(i, val - query(i, i));
16    }
17
18    T query(int i) {
19        T sum = 0;
20        for (i++; i > 0; i -= i & -i) sum += tree[i];
21        return sum;
22    }
23
24    T query(int l, int r) {
25        if (l > r) return 0;
26        return query(r) - query(l - 1);
27    }
28
29    // Returns first 0-based index where prefix sum >= val
30    int lower_bound(T val) {
31        int pos = 0;
32        for (int sz = 1 << (n ? __lg(n) : 0); sz > 0; sz >>= 1) {
33            if (pos + sz <= n && tree[pos + sz] < val) {
34                val -= tree[pos + sz];
35                pos += sz;
36            }
37        }
38        return pos;
39    }
40 };

```

2.2.2 FenwickTree2D

```

1 // 2D Fenwick Tree - 0-based indexing
2 // Space: O(N * M), Time: O(log N * log M) per update/query
3 template<typename T = ll>
4 struct FenwickTree2D {
5     int n, m;
6     vector<vector<T>> tree;
7
8     FenwickTree2D(int n = 0, int m = 0) : n(n), m(m), tree(n + 1, vector<T>(m + 1, 0)) {}
9
10    void add(int x, int y, T val) {
11        for (int i = x + 1; i <= n; i += i & -i) {
12            for (int j = y + 1; j <= m; j += j & -j) {
13                tree[i][j] += val;
14            }
15        }
16    }
17
18    void set(int x, int y, T val) {
19        add(x, y, val - query(x, y, x, y));
20    }
21 };

```

```

22 T query(int x, int y) {
23     T sum = 0;
24     for (int i = min(x + 1, n); i > 0; i -= i & -i) {
25         for (int j = min(y + 1, m); j > 0; j -= j & -j) {
26             sum += tree[i][j];
27         }
28     }
29     return sum;
30 }
31
32 // 2D Subgrid Range Sum from (x1, y1) to (x2, y2) inclusive
33 T query(int x1, int y1, int x2, int y2) {
34     if (x1 > x2 || y1 > y2) return 0;
35     return query(x2, y2) - query(x1 - 1, y2) - query(x2, y1 - 1) +
36         query(x1 - 1, y1 - 1);
37 };

```

2.2.3 Offline2DFenwickTree

```

1 // Offline 2D Fenwick Tree (Compressed 2D BIT)
2 // Space: O(N + Q log N), Time: O(log^2 N) per query
3 template<typename T = ll>
4 struct BIT2D {
5     int n;
6     vector<vector<int>>> vals;
7     vector<vector<T>>> bit;
8
9     int ind(const vector<int>& v, int x) {
10         return upper_bound(all(v), x) - v.begin() - 1;
11     }
12
13     // n: max row index, todo: all (r, c) update coordinates that will
14     // be called
15     BIT2D(int n, vector<array<int, 2>> todo) : n(n + 1), vals(n + 1),
16         bit(n + 1) {
17         sort(all(todo), [](const auto& a, const auto& b) { return a[1]
18             < b[1]; });
19
20         for (int i = 0; i <= n; i++) vals[i].pb(INT_MIN);
21         for (auto [r, c] : todo) {
22             for (int x = r; x <= n; x |= x + 1) {
23                 if (vals[x].back() != c) vals[x].pb(c);
24             }
25         }
26         for (int i = 0; i <= n; i++) bit[i].resize(vals[i].size(), 0);
27     }
28
29     void add(int r, int c, T val) {
30         for (; r <= n; r |= r + 1) {
31             for (int i = ind(vals[r], c); i < (int)bit[r].size(); i |=
32                 i + 1) {
33                 bit[r][i] += val;
34             }
35         }
36     }
37
38     T query(int r, int c) {
39         T res = 0;
40         for (; r >= 0; r = (r & (r + 1)) - 1) {
41             for (int i = ind(vals[r], c); i >= 0; i = (i & (i + 1)) -
42                 1) {
43                 res += bit[r][i];
44             }
45         }
46         return res;
47     }
48
49     T query(int r1, int c1, int r2, int c2) {
50         return query(r2, c2) - query(r2, c1 - 1) - query(r1 - 1, c2) +
51             query(r1 - 1, c1 - 1);
52     }
53
54     void reset() {
55         for (auto& v : bit) fill(all(v), 0);
56     }
57 };

```

2.3 Sparse Table

2.3.1 SparseTable

```

1 // Sparse Table (Range Minimum/Maximum Query) - O(N log N) build, O(1)
2 // query
3 template<typename T, typename F = function<T(T, T)>>
4 struct SparseTable {
5     int n;
6     vector<vector<T>>> mat;
7     F func;
8
9     SparseTable() = default;
10    SparseTable(const vector<T>& a, F f = [](T x, T y) { return max(x,
11        y); }) : n(a.size()), func(f) {
12        int k = n ? __lg(n) + 1 : 0;
13        mat.assign(k, a);
14        for (int j = 1; j < k; j++) {
15            for (int i = 0; i + (1 << j) <= n; i++) {
16                mat[j][i] = func(mat[j - 1][i], mat[j - 1][i + (1 << (j
17                    - 1))]);
18            }
19        }
20    }
21
22    T query(int l, int r) const {
23        int j = __lg(r - l + 1);
24        return func(mat[j][l], mat[j][r - (1 << j) + 1]);
25    }
26 };

```

2.3.2 SparseTable2D

```

1 // 2D Sparse Table - O(N * M * log N * log M) build, O(1) query
2 template<typename T = ll>

```

```

3 struct SparseTable2D {
4     int n, m;
5     // jmp[kr][kc][i][j] = max in 2^kr x 2^kc subgrid starting at (i, j)
6     vector<vector<vector<vector<T>>>>> jmp;
7
8     SparseTable2D(const vector<vector<T>>& grid) {
9         n = grid.size();
10        if (!n) return;
11        m = grid[0].size();
12        if (!m) return;
13
14        int log_n = __lg(n) + 1, log_m = __lg(m) + 1;
15        jmp.assign(log_n, vector<vector<vector<T>>>>(log_m, vector<
16            vector<T>>>(n, vector<T>(m))));
17
18        for (int i = 0; i < n; i++) {
19            for (int j = 0; j < m; j++) {
20                jmp[0][0][i][j] = grid[i][j];
21            }
22        }
23
24        for (int kc = 1; kc < log_m; kc++) {
25            for (int i = 0; i < n; i++) {
26                for (int j = 0; j + (1 << kc) <= m; j++) {
27                    jmp[0][kc][i][j] = max(jmp[0][kc - 1][i][j], jmp
28                        [0][kc - 1][i][j + (1 << (kc - 1))]);
29                }
30            }
31        }
32
33        for (int kr = 1; kr < log_n; kr++) {
34            for (int kc = 0; kc < log_m; kc++) {
35                for (int i = 0; i + (1 << kr) <= n; i++) {
36                    for (int j = 0; j + (1 << kc) <= m; j++) {
37                        jmp[kr][kc][i][j] = max(jmp[kr - 1][kc][i][j],
38                            jmp[kr - 1][kc][i + (1 << (kr - 1))][j]);
39                    }
40                }
41            }
42        }
43
44        T query(int r1, int c1, int r2, int c2) const {
45            int kr = __lg(r2 - r1 + 1), kc = __lg(c2 - c1 + 1);
46            return max({jmp[kr][kc][r1][c1],
47                jmp[kr][kc][r1][c2 - (1 << kc) + 1],
48                jmp[kr][kc][r2 - (1 << kr) + 1][c1],
49                jmp[kr][kc][r2 - (1 << kr) + 1][c2 - (1 << kc) +
50                    1]});
51        }
52    };
53 };

```

2.3.3 SquareSparseTable

```

1 // 2D Square Sparse Table - O(N * M * log(min(N, M))) build, O(1)
2 // square query
3 template<typename T = ll>
4 struct SquareSparseTable {
5     int n, m;
6     // jmp[k][i][j] = max of 2^k x 2^k square starting at (i, j)
7     vector<vector<vector<T>>>> jmp;
8
9     SquareSparseTable(const vector<vector<T>>& grid) {
10        n = grid.size();
11        if (!n) return;
12        m = grid[0].size();
13        if (!m) return;
14
15        int log_max = __lg(min(n, m)) + 1;
16        jmp.assign(log_max, vector<vector<T>>>(n, vector<T>(m)));
17
18        for (int i = 0; i < n; i++) {
19            for (int j = 0; j < m; j++) {
20                jmp[0][i][j] = grid[i][j];
21            }
22        }
23
24        for (int k = 1; k < log_max; k++) {
25            int pw = 1 << (k - 1);
26            for (int i = 0; i + (pw * 2) <= n; i++) {
27                for (int j = 0; j + (pw * 2) <= m; j++) {
28                    jmp[k][i][j] = max({jmp[k - 1][i][j],
29                        jmp[k - 1][i][j + pw],
30                        jmp[k - 1][i + pw][j],
31                        jmp[k - 1][i + pw][j + pw]});
32                }
33            }
34        }
35
36        // Queries max in square subgrid from (r1, c1) to (r2, c2)
37        T query(int r1, int c1, int r2, int c2) const {
38            int k = __lg(r2 - r1 + 1), pw = 1 << k;
39            return max({jmp[k][r1][c1],
40                jmp[k][r1][c2 - pw + 1],
41                jmp[k][r2 - pw + 1][c1],
42                jmp[k][r2 - pw + 1][c2 - pw + 1]});
43        }
44    };
45 };

```

2.4 Trie

2.4.1 StringTrie

```

1 // Prefix Trie for Strings
2 // Time: O(L) per insert/query, Space: O(N * L * Alphabet)
3 struct Trie {
4     vector<vector<int>>> nxt;
5     vector<int> cnt, end_cnt;
6     int alpha;
7
8     Trie(int alpha = 26) : alpha(alpha) {
9         add_node();
10    }

```

```

10 }
11
12 int add_node() {
13     nxt.emplace_back(alpha, -1);
14     cnt.emplace_back(0);
15     end_cnt.emplace_back(0);
16     return nxt.size() - 1;
17 }
18
19 void insert(const string& s, int val = 1) {
20     int u = 0;
21     cnt[u] += val;
22     for (char c : s) {
23         int ch = c - 'a';
24         if (nxt[u][ch] == -1) nxt[u][ch] = add_node();
25         u = nxt[u][ch];
26         cnt[u] += val;
27     }
28     end_cnt[u] += val;
29 }
30
31 int count_prefix(const string& s) {
32     int u = 0;
33     for (char c : s) {
34         int ch = c - 'a';
35         if (nxt[u][ch] == -1) return 0;
36         u = nxt[u][ch];
37     }
38     return cnt[u];
39 }
40 };

```

2.4.2 BinaryTrie

```

1 // Binary Trie for 64-bit integers and XOR operations
2 // Time: O(BITS) per operation
3 struct BinaryTrie {
4     vector<array<int, 2>> nxt;
5     vector<int> cnt;
6     int max_bit;
7
8     BinaryTrie(int max_bit = 60) : max_bit(max_bit) {
9         add_node();
10    }
11
12    int add_node() {
13        nxt.push_back({-1, -1});
14        cnt.push_back(0);
15        return nxt.size() - 1;
16    }
17
18    void insert(ll x, int val = 1) {
19        int u = 0;
20        cnt[u] += val;
21        for (int i = max_bit; i >= 0; i--) {
22            int b = (x >> i) & 1;
23            if (nxt[u][b] == -1) nxt[u][b] = add_node();
24            u = nxt[u][b];
25            cnt[u] += val;
26        }
27    }
28
29    // Returns element in trie maximizing (x ^ element)
30    ll max_xor(ll x) const {
31        if (!cnt[0]) return -1e18;
32        int u = 0;
33        ll res = 0;
34        for (int i = max_bit; i >= 0; i--) {
35            int b = (x >> i) & 1;
36            int want = b ^ 1;
37            if (nxt[u][want] != -1 && cnt[nxt[u][want]] > 0) {
38                res |= (1LL << i);
39                u = nxt[u][want];
40            } else {
41                u = nxt[u][b];
42            }
43        }
44        return res;
45    }
46 };

```

2.5 Disjoint Set Union (DSU)

2.5.1 DSU

```

1 // Disjoint Set Union (DSU) - Path Compression & Union by Size
2 // Time: O(alpha(N)) per operation
3 struct DSU {
4     vector<int> par, sz;
5
6     DSU(int n = 0) : par(n), sz(n, 1) {
7         iota(all(par), 0);
8     }
9
10    int find(int x) {
11        return par[x] == x ? x : par[x] = find(par[x]);
12    }
13
14    bool same(int x, int y) {
15        return find(x) == find(y);
16    }
17
18    bool join(int x, int y) {
19        x = find(x);
20        y = find(y);
21        if (x == y) return false;
22        if (sz[x] < sz[y]) swap(x, y);
23        sz[x] += sz[y];
24        par[y] = x;
25        return true;
26    }
27
28    int size(int x) {
29        return sz[find(x)];
30    }
31 };

```

```

30 }
31 };

```

2.5.2 DSURollback

```

1 // DSU with Rollback - No path compression to allow backtracking
2 // Time: O(log N) per operation, O(1) rollback
3 struct DSURollback {
4     vector<int> par, sz;
5     vector<pair<int, int>> history; // {child_node, old_parent_size}
6
7     DSURollback(int n = 0) : par(n), sz(n, 1) {
8         iota(all(par), 0);
9     }
10
11    int find(int x) {
12        return par[x] == x ? x : find(par[x]);
13    }
14
15    bool same(int x, int y) {
16        return find(x) == find(y);
17    }
18
19    bool join(int x, int y) {
20        x = find(x);
21        y = find(y);
22        if (x == y) {
23            history.eb(-1, -1);
24            return false;
25        }
26        if (sz[x] < sz[y]) swap(x, y);
27        history.eb(y, sz[x]);
28        sz[x] += sz[y];
29        par[y] = x;
30        return true;
31    }
32
33    int size(int x) {
34        return sz[find(x)];
35    }
36
37    void rollback() {
38        if (history.empty()) return;
39        auto [y, old_sz_x] = history.back();
40        history.pop_back();
41        if (y != -1) {
42            sz[par[y]] = old_sz_x;
43            par[y] = y;
44        }
45    }
46
47    int get_state() const {
48        return history.size();
49    }
50
51    void rollback_to(int state) {
52        while ((int)history.size() > state) rollback();
53    }
54 };

```

2.6 Square Root Decomposition

2.6.1 MoAlgorithm

```

1 class MoArray {
2 private:
3     struct Query {
4         int l, r, idx;
5         int bnd;
6
7         bool operator<(const Query& other) const {
8             if (bnd != other.bnd)
9                 return bnd < other.bnd;
10            return (bnd & 1) ? (r < other.r) : (r > other.r);
11        }
12    };
13
14    int n;
15    int block_size;
16    vector<int> a;
17
18    // --- PROBLEM SPECIFIC STATE ---
19    long long ans;
20    vector<int> freq;
21
22    inline void add(int idx) {
23        if (freq[a[idx]] == 0) ans++;
24        freq[a[idx]]++;
25    }
26
27    inline void remove(int idx) {
28        freq[a[idx]]--;
29        if (freq[a[idx]] == 0) ans--;
30    }
31
32    inline long long get_answer() const {
33        return ans;
34    }
35    // -----
36
37 public:
38     MoArray(const vector<int>& arr) : n(arr.size()), a(arr) {
39         int max_val = 0;
40         for (int x : a) {
41             max_val = max(max_val, x);
42         }
43         freq.assign(max_val + 1, 0);
44         ans = 0;
45     }
46
47     vector<long long> solve(const vector<pair<int, int>>& raw_queries) {
48         int q = raw_queries.size();
49         if (q == 0) return {};
50     }
51 };

```

```

50     block_size = max(1, (int)(n / sqrt(q)));
51     vector<Query> queries(q);
52
53     for (int i = 0; i < q; ++i) {
54         queries[i].l = raw_queries[i].first;
55         queries[i].r = raw_queries[i].second;
56         queries[i].idx = i;
57         queries[i].bnd = queries[i].l / block_size;
58     }
59
60     sort(queries.begin(), queries.end());
61
62     vector<long long> answers(q);
63     int cur_l = 0, cur_r = -1;
64
65     for (const Query& qry : queries) {
66         while (cur_l > qry.l) add(--cur_l);
67         while (cur_r < qry.r) add(++cur_r);
68         while (cur_l < qry.l) remove(cur_l++);
69         while (cur_r > qry.r) remove(cur_r--);
70         answers[qry.idx] = get_answer();
71     }
72
73     return answers;
74 }
75
76 };

```

2.6.2 MoOnSubtrees

```

1 class MoSubtree {
2 private:
3     struct Query {
4         int l, r, idx;
5         int bnd;
6
7         bool operator<(const Query& other) const {
8             if (bnd != other.bnd) return bnd < other.bnd;
9             return (bnd & 1) ? (r < other.r) : (r > other.r);
10        }
11    };
12
13    int n;
14    int block_size;
15    vector<vector<int>> adj;
16    vector<int> in, out, flat, clr;
17
18    // --- PROBLEM SPECIFIC STATE ---
19    int mx_freq;
20    vector<int> freq;
21    vector<long long> sum_freq;
22
23    inline void add(int node_idx) {
24        int c = clr[node_idx];
25        sum_freq[freq[c]] -= c;
26        freq[c]++;
27        sum_freq[freq[c]] += c;
28        if (freq[c] > mx_freq) mx_freq = freq[c];
29    }
30
31    inline void remove(int node_idx) {
32        int c = clr[node_idx];
33        sum_freq[freq[c]] -= c;
34        freq[c]--;
35        sum_freq[freq[c]] += c;
36        while (mx_freq > 0 && sum_freq[mx_freq] == 0) mx_freq--;
37    }
38
39    inline long long get_answer() const {
40        return sum_freq[mx_freq];
41    }
42    // -----
43
44    void dfs(int u, int p) {
45        in[u] = flat.size();
46        flat.push_back(u);
47        for (int v : adj[u]) {
48            if (v == p) continue;
49            dfs(v, u);
50        }
51        out[u] = flat.size() - 1;
52    }
53
54 public:
55    MoSubtree(int n, int max_val, const vector<int>& colors)
56        : n(n), adj(n + 1), in(n + 1), out(n + 1), clr(colors) {
57        mx_freq = 0;
58        freq.assign(max_val + 1, 0);
59        sum_freq.assign(n + 1, 0);
60    }
61
62    void add_edge(int u, int v) {
63        adj[u].push_back(v);
64        adj[v].push_back(u);
65    }
66
67    vector<long long> solve(int root, const vector<int>& query_nodes) {
68        flat.reserve(n);
69        dfs(root, 0);
70
71        int q = query_nodes.size();
72        if (q == 0) return {};
73
74        block_size = max(1, (int)(n / sqrt(q)));
75        vector<Query> queries(q);
76
77        for (int i = 0; i < q; ++i) {
78            int u = query_nodes[i];
79            queries[i].l = in[u];
80            queries[i].r = out[u];
81            queries[i].idx = i;
82            queries[i].bnd = queries[i].l / block_size;
83        }
84    }

```

```

85        sort(queries.begin(), queries.end());
86
87        vector<long long> answers(q);
88        int cur_l = 0, cur_r = -1;
89
90        for (const Query& qry : queries) {
91            while (cur_l > qry.l) add(flat[--cur_l]);
92            while (cur_r < qry.r) add(flat[++cur_r]);
93            while (cur_l < qry.l) remove(flat[cur_l++]);
94            while (cur_r > qry.r) remove(flat[cur_r--]);
95            answers[qry.idx] = get_answer();
96        }
97
98        return answers;
99    }
100 };

```

2.6.3 MoOnPaths

```

1 class MoTreePath {
2 private:
3     int n, LOG, block_size;
4     vector<vector<int>> adj, anc;
5     vector<int> depth, in, out, flat;
6     vector<long long> up, a;
7     vector<bool> exist;
8
9     // --- PROBLEM SPECIFIC STATE ---
10    long long ans;
11    vector<deque<int>> dep; // Tracks nodes by depth
12
13    void add(int idx) {
14        if (!dep[depth[idx]].empty()) {
15            ans += 1ll * a[dep[depth[idx]].back()] * a[idx];
16        }
17        dep[depth[idx]].push_back(idx);
18    }
19
20    void remove(int idx) {
21        if (dep[depth[idx]].size() == 2) {
22            ans -= a[*dep[depth[idx]].begin()] * a[*next(dep[depth[idx]].begin())];
23        }
24        if (dep[depth[idx]].front() == idx) {
25            dep[depth[idx]].pop_front();
26        } else {
27            dep[depth[idx]].pop_back();
28        }
29    }
30
31    void update(int idx) {
32        int node = flat[idx];
33        if (exist[node]) {
34            remove(node);
35        } else {
36            add(node);
37        }
38        exist[node] = !exist[node];
39    }
40
41    inline long long get_answer() const {
42        return ans;
43    }
44    // -----
45
46    void dfs(int u, int p = -1) {
47        if (p != -1) {
48            anc[u][0] = p;
49            up[u] = a[u] * a[u] + up[p];
50        } else {
51            depth[u] = 0;
52            up[u] = a[u] * a[u];
53        }
54        in[u] = flat.size();
55        flat.push_back(u);
56
57        for (int v : adj[u]) {
58            if (v == p) continue;
59            depth[v] = depth[u] + 1;
60            dfs(v, u);
61        }
62
63        out[u] = flat.size();
64        flat.push_back(u); // Push again for Eulerian tour
65    }
66
67    void build_lca() {
68        for (int k = 1; k < LOG; k++) {
69            for (int i = 1; i <= n; i++) {
70                anc[i][k] = anc[anc[i][k - 1]][k - 1];
71            }
72        }
73    }
74
75    int lift(int x, int k) {
76        for (int bit = 0; bit < LOG; bit++) {
77            if ((1 << bit) & k) x = anc[x][bit];
78        }
79        return x;
80    }
81
82    int lca(int u, int v) {
83        if (depth[u] < depth[v]) swap(u, v);
84        u = lift(u, depth[u] - depth[v]);
85        if (u == v) return u;
86        for (int bit = LOG - 1; bit >= 0; bit--) {
87            if (anc[u][bit] != anc[v][bit]) {
88                u = anc[u][bit];
89                v = anc[v][bit];
90            }
91        }
92        return anc[u][0];
93    }
94 }

```

```

95 public:
96     struct Query {
97         int l, r, idx, x = -1;
98         int bnd;
99
100         bool operator<(const Query& other) const {
101             if (bnd != other.bnd) return bnd < other.bnd;
102             return (bnd & 1) ? (r < other.r) : (r > other.r);
103         }
104     };
105
106     MoTreePath(int n_nodes, const vector<long long>& node_values)
107         : n(n_nodes), a(node_values), LOG(log2(n_nodes) + 2) {
108
109         adj.assign(n + 1, vector<int>());
110         anc.assign(n + 1, vector<int>(LOG, 0));
111         depth.assign(n + 1, 0);
112         in.assign(n + 1, 0);
113         out.assign(n + 1, 0);
114         up.assign(n + 1, 0);
115         exist.assign(n + 1, false);
116         dep.assign(n + 1, deque<int>());
117         ans = 0;
118     }
119
120     void add_edge(int u, int v) {
121         adj[u].push_back(v);
122         adj[v].push_back(u);
123     }
124
125     vector<long long> solve(int root, vector<Query>& queries) {
126         flat.clear();
127         dfs(root);
128         build_lca();
129
130         int q = queries.size();
131         if (q == 0) return {};
132
133         block_size = max(1, (int)(flat.size() / sqrt(q)));
134
135         for (auto& qry : queries) {
136             qry.bnd = qry.l / block_size;
137         }
138
139         sort(queries.begin(), queries.end());
140
141         vector<long long> answers(q);
142         int cur_l = 0, cur_r = -1;
143
144         for (const Query& qry : queries) {
145             while (cur_l < qry.l) update(cur_l++);
146             while (cur_r > qry.r) update(cur_r--);
147             while (cur_l > qry.l) update(--cur_l);
148             while (cur_r < qry.r) update(++cur_r);
149
150             answers[qry.idx] = get_answer() + (qry.x != -1 ? up[qry.x] : 0);
151         }
152         return answers;
153     }
154 };

```

2.7 Balanced Binary Search Tree (BBST)

2.7.1 Treap

```

1 mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
2
3 template <typename T>
4 struct Treap {
5     struct Node {
6         T key, val, sum;
7         T lazy = 0; // 1. Added lazy tag
8         int sz = 1;
9         uint64_t p = rng();
10        Node *l = nullptr, *r = nullptr;
11
12        Node(T k, T v = T()) : key(k), val(v), sum(v) {}
13    } * root = nullptr;
14
15    ~Treap() { destroy(root); }
16
17    void destroy(Node* t) {
18        if (!t) return;
19        destroy(t->l);
20        destroy(t->r);
21        delete t;
22    }
23
24    int sz(Node* t) { return t ? t->sz : 0; }
25    T sub_val(Node* t) { return t ? t->sum : 0; }
26
27    // 2. The Push Function: Propagates pending updates to children
28    void push(Node* t) {
29        if (!t || t->lazy == 0) return;
30
31        if (t->l) {
32            t->l->key += t->lazy; // Shift the key (critical for the DP
33                               // solution)
34            t->l->val += t->lazy; // Shift the value
35            t->l->sum += t->lazy * sz(t->l); // Update subtree sum
36            t->l->lazy += t->lazy; // Pass the tag down
37        }
38        if (t->r) {
39            t->r->key += t->lazy;
40            t->r->val += t->lazy;
41            t->r->sum += t->lazy * sz(t->r);
42            t->r->lazy += t->lazy;
43        }
44        t->lazy = 0; // Clear the tag on the current node
45    }
46
47    Node* update(Node* t) {
48        if (!t) return nullptr;
49        push(t);
50        if (t->l->p > t->r->p) {
51            rotate_right(t);
52        } else if (t->r->p > t->l->p) {
53            rotate_left(t);
54        }
55        return update(t);
56    }
57
58    void rotate_left(Node* t) {
59        Node* r = t->r;
60        t->r = r->l;
61        r->l = t;
62        update(t);
63        update(r);
64    }
65
66    void rotate_right(Node* t) {
67        Node* l = t->l;
68        t->l = l->r;
69        l->r = t;
70        update(t);
71        update(l);
72    }
73
74    Node* insert(Node* t, Node* x) {
75        if (!t) return x;
76        push(t);
77        if (x->key < t->key) {
78            t->l = insert(t->l, x);
79        } else {
80            t->r = insert(t->r, x);
81        }
82        return t;
83    }
84
85    Node* erase(Node* t, Node* x) {
86        if (!t) return nullptr;
87        push(t);
88        if (x->key < t->key) {
89            t->l = erase(t->l, x);
90        } else if (x->key > t->key) {
91            t->r = erase(t->r, x);
92        } else {
93            if (!t->l) return t->r;
94            if (!t->r) return t->l;
95            Node* l = t->l;
96            Node* r = t->r;
97            t->l = l->r;
98            t->r = r->l;
99            l->r = t;
100            r->l = t;
101            update(t);
102            update(l);
103            update(r);
104        }
105        return t;
106    }
107
108    Node* find_by_order(Node* t, int k) {
109        if (!t) return nullptr;
110        push(t);
111        int left_sz = sz(t->l);
112        if (k < left_sz) return find_by_order(t->l, k);
113        if (k == left_sz) return t;
114        if (k > left_sz + 1) return find_by_order(t->r, k - left_sz - 1);
115    }
116
117    Node* find_by_order(int k) { return find_by_order(root, k); }
118
119    int order_of_key(Node* t, T k) {
120        if (!t) return 0;
121        push(t);
122        if (k < t->key) return order_of_key(t->l, k);
123        if (k >= t->key) return 1 + order_of_key(t->r, k);
124    }
125
126    int order_of_key(T k) { return order_of_key(root, k); }
127
128    void print(Node* t) {
129        if (!t) return;
130        push(t);
131        print(t->l);
132        cout << t->key << " ";
133        print(t->r);
134    }
135
136    void print() {
137        print(root);
138        cout << '\n';
139    }
140
141    ~Treap() { destroy(root); }
142 };

```

```

48         t->sz = 1 + sz(t->l) + sz(t->r);
49         t->sum = t->val + sub_val(t->l) + sub_val(t->r);
50     }
51     return t;
52 }
53
54 Node* merge(Node* l, Node* r) {
55     push(l); // 3. Push before making routing decisions
56     push(r);
57     if (!l || !r) return l ? l : r;
58     if (l->p > r->p) {
59         l->r = merge(l->r, r);
60         return update(l);
61     }
62     r->l = merge(l, r->l);
63     return update(r);
64 }
65
66 pair<Node*, Node*> split(Node* t, T k) {
67     if (!t) return {nullptr, nullptr};
68     push(t); // 3. Push before evaluating keys for splits
69     if (t->key <= k) {
70         auto [l, r] = split(t->r, k);
71         t->r = l;
72         return {update(t), r};
73     }
74     auto [l, r] = split(t->l, k);
75     t->l = r;
76     return {l, update(t)};
77 }
78
79 // --- Core Operations ---
80
81 // Range Update Method
82 void add_range(T l_val, T r_val, T add_val) {
83     if (l_val > r_val) return;
84
85     auto [left_mid, right] = split(root, r_val);
86     auto [left, mid] = split(left_mid, l_val - 1);
87
88     // Apply the lazy tag to the entire isolated segment
89     if (mid) {
90         mid->key += add_val;
91         mid->val += add_val;
92         mid->sum += add_val * sz(mid);
93         mid->lazy += add_val;
94     }
95
96     root = merge(merge(left, mid), right);
97 }
98
99 bool search(T x) {
100     Node* curr = root;
101     while (curr) {
102         push(curr); // Push during traversal
103         if (curr->key == x) return true;
104         curr = (x < curr->key) ? curr->l : curr->r;
105     }
106     return false;
107 }
108
109 void insert(T x, T val = T()) {
110     auto [l, r] = split(root, x);
111     auto [l2, r2] = split(l, x - 1);
112     if (!r2) {
113         r2 = new Node(x, val);
114     } else {
115         push(r2); // Push before modifying
116         r2->val += val;
117         update(r2);
118     }
119     root = merge(merge(l2, r2), r);
120 }
121
122 void erase(T x) {
123     auto [l, r] = split(root, x);
124     auto [l2, r2] = split(l, x - 1);
125     destroy(r2);
126     root = merge(l2, r);
127 }
128
129 // --- Utility Operations ---
130
131 Node* find_by_order(Node* t, int k) {
132     if (!t) return nullptr;
133     push(t); // Push before accessing children sizes
134     int left_sz = sz(t->l);
135     if (k < left_sz) return find_by_order(t->l, k);
136     if (k == left_sz) return t;
137     if (k > left_sz + 1) return find_by_order(t->r, k - left_sz - 1);
138 }
139
140 Node* find_by_order(int k) { return find_by_order(root, k); }
141
142 int order_of_key(Node* t, T k) {
143     if (!t) return 0;
144     push(t); // Push before comparing keys
145     if (t->key >= k) return order_of_key(t->l, k);
146     return sz(t->l) + 1 + order_of_key(t->r, k);
147 }
148
149 int order_of_key(T k) { return order_of_key(root, k); }
150
151 void print(Node* t) {
152     if (!t) return;
153     push(t); // Push before printing to get true values
154     print(t->l);
155     cout << t->key << " ";
156     print(t->r);
157 }
158
159 void print() {
160     print(root);
161     cout << '\n';
162 }
163
164 ~Treap() { destroy(root); }
165 };

```

2.7.2 ImplicitTreap

```
1 mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
2
3 template <typename T>
4 struct ImplicitTreap {
5     struct Node {
6         T val, sum;
7         T lazy = 0; // Lazy tag for range additions
8         bool rev = false; // Lazy tag for range reversals
9         int sz = 1;
10        uint64_t p = rng();
11        Node *l = nullptr, *r = nullptr;
12
13        Node(T v = T()) : val(v), sum(v) {}
14    } * root = nullptr;
15
16    // Default Constructor
17    ImplicitTreap() = default;
18
19    // Vector Constructor - Builds treap in O(N) time
20    ImplicitTreap(const vector<T>& a) {
21        build(a);
22    }
23
24    ~ImplicitTreap() { destroy(root); }
25
26    void destroy(Node* t) {
27        if (!t) return;
28        destroy(t->l);
29        destroy(t->r);
30        delete t;
31    }
32
33    int sz(Node* t) { return t ? t->sz : 0; }
34    T sub_val(Node* t) { return t ? t->sum : 0; }
35
36    // Propagates pending lazy updates to children
37    void push(Node* t) {
38        if (!t) return;
39
40        // 1. Process pending reversal
41        if (t->rev) {
42            swap(t->l, t->r);
43            if (t->l) t->l->rev ^= true;
44            if (t->r) t->r->rev ^= true;
45            t->rev = false;
46        }
47
48        // 2. Process pending value addition
49        if (t->lazy != 0) {
50            if (t->l) {
51                t->l->val += t->lazy;
52                t->l->sum += t->lazy * sz(t->l);
53                t->l->lazy += t->lazy;
54            }
55            if (t->r) {
56                t->r->val += t->lazy;
57                t->r->sum += t->lazy * sz(t->r);
58                t->r->lazy += t->lazy;
59            }
60            t->lazy = 0; // Clear the tag
61        }
62    }
63
64    Node* update(Node* t) {
65        if (t) {
66            t->sz = 1 + sz(t->l) + sz(t->r);
67            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
68        }
69        return t;
70    }
71
72    // --- O(N) Linear Time Build ---
73    void build(const vector<T>& a) {
74        destroy(root);
75        root = nullptr;
76        if (a.empty()) return;
77
78        vector<Node*> st;
79        for (const T& val : a) {
80            Node* curr = new Node(val);
81            Node* last = nullptr;
82
83            // Maintain max-heap property on random priority 'p'
84            while (!st.empty() && st.back()->p < curr->p) {
85                last = st.back();
86                st.pop_back();
87            }
88
89            curr->l = last;
90            if (!st.empty()) {
91                st.back()->r = curr;
92            }
93            st.push_back(curr);
94        }
95
96        // Post-order pass to compute subtree sizes and sums in O(N)
97        auto recompute = [&](auto& self, Node* t) -> void {
98            if (!t) return;
99            self(self, t->l);
100            self(self, t->r);
101            update(t);
102        };
103
104        root = st.front(); // Bottom of stack is the root
105        recompute(recompute, root);
106    }
107
108    // --- Dynamic Array Operations ---
109
110    Node* merge(Node* l, Node* r) {
111        push(l);
112        push(r);
```

```
113        if (!l || !r) return l ? l : r;
114        if (l->p > r->p) {
115            l->r = merge(l->r, r);
116            return update(l);
117        }
118        r->l = merge(l, r->l);
119        return update(r);
120    }
121
122    pair<Node*, Node*> split(Node* t, int k) {
123        if (!t) return {nullptr, nullptr};
124        push(t);
125
126        int left_sz = sz(t->l);
127        if (left_sz < k) {
128            auto [l, r] = split(t->r, k - left_sz - 1);
129            t->r = l;
130            return {update(t), r};
131        } else {
132            auto [l, r] = split(t->l, k);
133            t->l = r;
134            return {l, update(t)};
135        }
136    }
137
138    int size() { return sz(root); }
139
140    void insert(int idx, T val) {
141        auto [l, r] = split(root, idx);
142        Node* node = new Node(val);
143        root = merge(merge(l, node), r);
144    }
145
146    void erase(int idx) {
147        if (idx < 0 || idx >= sz(root)) return;
148        auto [left_mid, right] = split(root, idx + 1);
149        auto [left, mid] = split(left_mid, idx);
150        destroy(mid);
151        root = merge(left, right);
152    }
153
154    void reverse_range(int l, int r) {
155        if (l >= r || l < 0 || r >= sz(root)) return;
156
157        auto [left_mid, right] = split(root, r + 1);
158        auto [left, mid] = split(left_mid, l);
159
160        if (mid) mid->rev ^= true;
161
162        root = merge(merge(left, mid), right);
163    }
164
165    void add_range(int l, int r, T add_val) {
166        if (l > r || l < 0 || r >= sz(root)) return;
167
168        auto [left_mid, right] = split(root, r + 1);
169        auto [left, mid] = split(left_mid, l);
170
171        if (mid) {
172            mid->val += add_val;
173            mid->sum += add_val * sz(mid);
174            mid->lazy += add_val;
175        }
176
177        root = merge(merge(left, mid), right);
178    }
179
180    T query_range(int l, int r) {
181        if (l > r || l < 0 || r >= sz(root)) return T();
182
183        auto [left_mid, right] = split(root, r + 1);
184        auto [left, mid] = split(left_mid, l);
185
186        T ans = sub_val(mid);
187
188        root = merge(merge(left, mid), right);
189        return ans;
190    }
191
192    T get(int idx) {
193        return query_range(idx, idx);
194    }
195
196    void print(Node* t) {
197        if (!t) return;
198        push(t);
199        print(t->l);
200        cout << t->val << " ";
201        print(t->r);
202    }
203
204    void print() {
205        print(root);
206        cout << '\n';
207    }
208 };
```

2.8 Policy Based Data Structures

2.8.1 OrderedSet

```
1 // Policy-Based Data Structure: ordered_set
2 // Time: O(log N) per operation
3 #include <ext/pb_ds/assoc_container.hpp>
4 #include <ext/pb_ds/tree_policy.hpp>
5
6 using namespace __gnu_pbds;
7 template<typename T>
8 using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
9     tree_order_statistics_node_update>;
10
11 // Usage:
12 // st.find_by_order(k): Iterator to k-th element (0-based)
13 // st.order_of_key(x): Number of elements strictly smaller than x
```

2.9 Monotonic Data Structures

2.9.1 TwoStackQueue

```
1 struct TwoStackQ {
2     struct Node {
3         ll mx = -llinf, mn = llinf, val;
4
5         Node(): val(0) {
6             }
7
8         Node(ll x): mx(x), mn(x), val(x) {
9             }
10    };
11
12    stack<Node> a, b;
13
14    int size() { return a.size() + b.size(); }
15
16    void mrg(Node& a, Node& b) {
17        a.mn = min(a.mn, b.mn);
18        a.mx = max(a.mx, b.mx);
19    }
20
21    void push(ll val) {
22        auto nd = Node(val);
23        if (!a.empty()) mrg(nd, a.top());
24        a.push(nd);
25    }
26
27    void move() {
28        while (!a.empty()) {
29            auto nd = Node(a.top().val);
30            if (!b.empty()) mrg(nd, b.top());
31            b.push(nd), a.pop();
32        }
33    }
34
35    Node get() {
36        Node res;
37        if (!b.empty()) mrg(res, b.top());
38        if (!a.empty()) mrg(res, a.top());
39        return res;
40    }
41
42    void pop() {
43        if (b.empty()) move();
44        if (!b.empty()) b.pop();
45    }
46};
```

3 Graph Theory

3.1 Graph Traversal

3.1.1 TopoSort

```
1 // Topological Sort (Kahn's Algorithm) - O(V + E)
2 struct TopoSort {
3     int n;
4     vector<vector<int>> adj;
5     vector<int> deg, order;
6
7     TopoSort(int n = 0) : n(n), adj(n + 1), deg(n + 1, 0) {}
8
9     void add_edge(int u, int v) {
10         adj[u].pb(v);
11         deg[v]++;
12     }
13
14     bool solve() {
15         queue<int> q;
16         for (int i = 1; i <= n; i++) {
17             if (!deg[i]) q.push(i);
18         }
19         while (!q.empty()) {
20             int u = q.front();
21             q.pop();
22             order.pb(u);
23             for (int v : adj[u]) {
24                 if (--deg[v] == 0) q.push(v);
25             }
26         }
27         return (int)order.size() == n; // false if graph has cycles
28     }
29};
```

3.2 Shortest Paths

3.2.1 Dijkstra

```
1 // Dijkstra's Shortest Path Algorithm - O((V + E) log V)
2 struct Dijkstra {
3     int n;
4     vector<vector<pair<int, ll>>> adj;
5     vector<ll> dist;
6     vector<int> par;
7
8     Dijkstra(int n = 0) : n(n), adj(n + 1), dist(n + 1, llinf), par(n + 1, -1) {}
9
10    void add_edge(int u, int v, ll w, bool directed = false) {
11        adj[u].eb(v, w);
12        if (!directed) adj[v].eb(u, w);
13    }
14
15    void solve(int src) {
16        fill(all(dist), llinf);
17        fill(all(par), -1);
18        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>> pq;
19
20        dist[src] = 0;
21        pq.emplace(0, src);
```

```
22
23        while (!pq.empty()) {
24            auto [d, u] = pq.top();
25            pq.pop();
26            if (d > dist[u]) continue;
27
28            for (auto& [v, w] : adj[u]) {
29                if (dist[u] + w < dist[v]) {
30                    dist[v] = dist[u] + w;
31                    par[v] = u;
32                    pq.emplace(dist[v], v);
33                }
34            }
35        }
36    }
37};
```

3.2.2 FloydWarshall

```
1 // Floyd-Warshall All-Pairs Shortest Path - O(V^3)
2 struct FloydWarshall {
3     int n;
4     vector<vector<ll>> dist;
5
6     FloydWarshall(int n = 0) : n(n), dist(n + 1, vector<ll>(n + 1, llinf)) {}
7     for (int i = 0; i <= n; i++) dist[i][i] = 0;
8
9
10    void add_edge(int u, int v, ll w, bool directed = false) {
11        dist[u][v] = min(dist[u][v], w);
12        if (!directed) dist[v][u] = min(dist[v][u], w);
13    }
14
15    void solve() {
16        for (int k = 1; k <= n; k++) {
17            for (int i = 1; i <= n; i++) {
18                for (int j = 1; j <= n; j++) {
19                    if (dist[i][k] < llinf && dist[k][j] < llinf) {
20                        dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
21                    }
22                }
23            }
24        }
25    }
26};
```

3.3 Graph Connectivity

3.3.1 Bridges

```
1 // Bridges (Cut Edges) in Undirected Graph - O(V + E)
2 struct Bridges {
3     int n, timer;
4     vector<vector<int>> adj;
5     vector<int> tin, low;
6     vector<pair<int, int>> bridges;
7
8     Bridges(int n = 0) : n(n), timer(0), adj(n + 1), tin(n + 1, 0), low(n + 1, 0) {}
9
10    void add_edge(int u, int v) {
11        adj[u].pb(v);
12        adj[v].pb(u);
13    }
14
15    void dfs(int u, int p = 0) {
16        tin[u] = low[u] = ++timer;
17        for (int v : adj[u]) {
18            if (v == p) continue;
19            if (tin[v]) {
20                low[u] = min(low[u], tin[v]);
21            } else {
22                dfs(v, u);
23                low[u] = min(low[u], low[v]);
24                if (low[v] > tin[u]) bridges.eb(u, v);
25            }
26        }
27    }
28
29    void solve() {
30        timer = 0;
31        fill(all(tin), 0);
32        bridges.clear();
33        for (int i = 1; i <= n; i++) {
34            if (!tin[i]) dfs(i);
35        }
36    }
37};
```

3.3.2 CutPoints

```
1 // Articulation Points (Cut Vertices) in Undirected Graph - O(V + E)
2 struct ArticulationPoints {
3     int n, timer;
4     vector<vector<int>> adj;
5     vector<int> tin, low;
6     vector<bool> is_cut;
7
8     ArticulationPoints(int n = 0) : n(n), timer(0), adj(n + 1), tin(n + 1, 0), low(n + 1, 0), is_cut(n + 1, false) {}
9
10    void add_edge(int u, int v) {
11        adj[u].pb(v);
12        adj[v].pb(u);
13    }
14
15    void dfs(int u, int p = 0) {
16        tin[u] = low[u] = ++timer;
17        int children = 0;
18        for (int v : adj[u]) {
19            if (v == p) continue;
```

```

20         if (tin[v]) {
21             low[u] = min(low[u], tin[v]);
22         } else {
23             dfs(v, u);
24             low[u] = min(low[u], low[v]);
25             if (low[v] >= tin[u] && p != 0) is_cut[u] = true;
26             children++;
27         }
28     }
29     if (p == 0 && children > 1) is_cut[u] = true;
30 }
31
32 void solve() {
33     timer = 0;
34     fill(all(tin), 0);
35     fill(all(is_cut), false);
36     for (int i = 1; i <= n; i++) {
37         if (!tin[i]) dfs(i);
38     }
39 }
40 };

```

3.3.3 SCC

```

1 // Strongly Connected Components (Tarjan's Algorithm) - O(V + E)
2 struct SCC {
3     int n, timer = 0, num_comps = 0;
4     vector<vector<int>> adj;
5     vector<int> tin, low, comp;
6     vector<bool> in_st;
7     stack<int> st;
8     vector<vector<int>> comps;
9
10    SCC(int n = 0) : n(n), adj(n + 1), tin(n + 1, 0), low(n + 1, 0),
11        comp(n + 1, -1), in_st(n + 1, false) {}
12
13    void add_edge(int u, int v) {
14        adj[u].pb(v);
15    }
16
17    void dfs(int u) {
18        tin[u] = low[u] = ++timer;
19        st.push(u);
20        in_st[u] = true;
21
22        for (int v : adj[u]) {
23            if (!tin[v]) {
24                dfs(v);
25                low[u] = min(low[u], low[v]);
26            } else if (in_st[v]) {
27                low[u] = min(low[u], tin[v]);
28            }
29        }
30
31        if (low[u] == tin[u]) {
32            comps.pb({});
33            while (true) {
34                int v = st.top();
35                st.pop();
36                in_st[v] = false;
37                comp[v] = num_comps;
38                comps.back().pb(v);
39                if (u == v) break;
40            }
41            num_comps++;
42        }
43
44    void solve() {
45        for (int i = 1; i <= n; i++) {
46            if (!tin[i]) dfs(i);
47        }
48    }
49
50    vector<vector<int>> build_dag() {
51        vector<vector<int>> dag(num_comps);
52        for (int u = 1; u <= n; u++) {
53            for (int v : adj[u]) {
54                if (comp[u] != comp[v]) dag[comp[u]].pb(comp[v]);
55            }
56        }
57        return dag;
58    }
59 };

```

3.4 Satisfiability

3.4.1 2SAT

```

1 struct TwoSat {
2     int N;
3     vector<pair<int, int>> edges;
4
5     TwoSat(int _N = 0) {
6         init(_N);
7     }
8
9     void init(int _N = 0) {
10         N = _N;
11     }
12
13     int addVar() { return N++; }
14
15     // x or y, edges will be refined in the end
16     void either(int x, int y) {
17         x = max(2 * x, -1 - 2 * x);
18         y = max(2 * y, -1 - 2 * y);
19         edges.pb({x, y});
20     }
21
22     void implies(int x, int y) {
23         either(-x, y);
24     }
25 }

```

```

26 void must(int x) {
27     either(x, x);
28 }
29
30 void XOR(int x, int y) {
31     either(x, y);
32     either(-x, -y);
33 }
34
35 vector<bool> solve(int _N = -1) {
36     if (_N != -1) N = _N;
37     SCC scc;
38     scc.init(2 * N);
39     for (auto e: edges) {
40         scc.addEdge(e.st ^ 1, e.nd);
41         scc.addEdge(e.nd ^ 1, e.st);
42     }
43     scc.gen();
44     for (int i = 0; i < 2 * N; ++i) {
45         if (scc.compOf[i] == scc.compOf[i ^ 1]) return {};
46     }
47     vector<vector<int>> comps = scc.comps;
48     reverse(all(comps));
49     vector<int> compOf(2 * N);
50     for (int i = 0; i < comps.size(); ++i) {
51         for (auto e: comps[i])
52             compOf[e] = i;
53     }
54     vector<int> tmp(comps.size());
55     for (int i = 0; i < comps.size(); ++i) {
56         if (!tmp[i]) {
57             tmp[i] = 1;
58             for (auto e: comps[i])
59                 tmp[compOf[e ^ 1]] = -1;
60         }
61     }
62     vector<bool> ans(N);
63     for (int i = 0; i < N; ++i)
64         ans[i] = tmp[compOf[2 * i]] == 1;
65     return ans;
66 }
67
68 void atMostOne(const vector<int> &li) {
69     if (li.size() <= 1) return;
70     int last = -li[1];
71     for (int i = 2; i < li.size(); i++) {
72         int next = addVar();
73         implies(li[i], last);
74         either(last, next);
75         implies(li[i], next);
76         last = -next;
77     }
78     implies(li[0], last);
79 }
80 };

```

3.5 Eulerian Path

3.5.1 EulerianPath

```

1 struct EulerianPath {
2     int n;
3     vector<vector<pair<int, int>>> adj; // {neighbor, edge_index}
4     vector<bool> used_edge;
5     vector<int> in_deg, out_deg;
6     vector<int> path;
7
8     EulerianPath(int n, int m) : n(n), adj(n + 1), used_edge(m, false),
9         in_deg(n + 1, 0), out_deg(n + 1, 0) {}
10
11     void add_edge(int u, int v, int edge_idx, bool directed = true) {
12         adj[u].push_back({v, edge_idx});
13         out_deg[u]++; in_deg[v]++;
14         if (!directed) {
15             adj[v].push_back({u, edge_idx});
16             out_deg[v]++; in_deg[u]++;
17         }
18     }
19
20     bool solve(int start_node) {
21         vector<int> ptr(n + 1, 0);
22         stack<int> st;
23         st.push(start_node);
24
25         while (!st.empty()) {
26             int u = st.top();
27             if (ptr[u] < adj[u].size()) {
28                 auto [v, id] = adj[u][ptr[u]++];
29                 if (!used_edge[id]) {
30                     used_edge[id] = true;
31                     st.push(v);
32                 }
33             } else {
34                 path.push_back(u);
35                 st.pop();
36             }
37         }
38         reverse(path.begin(), path.end());
39
40         // Check if all edges were visited
41         for (bool used : used_edge) {
42             if (!used) return false;
43         }
44         return true;
45     };
46 };

```

3.6 Flow and Min Cut

3.6.1 Dinic

```

1 struct Dinic {
2     struct Edge {
3         int to, rev;

```

```

4         ll c, oc;
5         ll flow() { return max(oc - c, 0LL); } // if you need flows
6     };
7
8     vector<int> lvl, ptr, q;
9     vector<vector<Edge>> > adj;
10
11     Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {
12     }
13
14     void addEdge(int a, int b, ll c, ll rcap = 0) {
15         adj[a].push_back({b, (int) adj[b].size(), c, c});
16         adj[b].push_back({a, (int) adj[a].size() - 1, rcap, rcap});
17     }
18
19     ll dfs(int v, int t, ll f) {
20         if (v == t || !f) return f;
21         for (int &i = ptr[v]; i < (int) adj[v].size(); i++) {
22             Edge &e = adj[v][i];
23             if (lvl[e.to] == lvl[v] + 1)
24                 if (ll p = dfs(e.to, t, min(f, e.c))) {
25                     e.c -= p, adj[e.to][e.rev].c += p;
26                     return p;
27                 }
28         }
29         return 0;
30     }
31
32     ll calc(int s, int t) {
33         ll flow = 0;
34         q[0] = s;
35         for (int L = 0; L < 31; L++)
36             do {
37                 // 'int L=30' maybe faster for random data
38                 lvl = ptr = vector<int>((int) q.size());
39                 int qi = 0, qe = lvl[s] = 1;
40                 while (qi < qe && !lvl[t]) {
41                     int v = q[qi++];
42                     for (Edge e: adj[v])
43                         if (!lvl[e.to] && e.c >> (30 - L))
44                             q[qe++] = e.to, lvl[e.to] = lvl[v] + 1;
45                 }
46                 while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
47             } while (lvl[t]);
48         return flow;
49     }
50
51     bool leftOfMinCut(int a) { return lvl[a] != 0; }
52 };

```

3.6.2 MCMF

```

1 struct Edge {
2     int to;
3     int cost;
4     int cap, flow, backEdge;
5 };
6
7 struct MCMF
8 {
9
10     const int inf = 1000000010;
11     int n;
12     vector<vector<Edge>> g;
13
14     MCMF(int _n) {
15         n = _n + 1;
16         g.resize(n);
17     }
18
19     void addEdge(int u, int v, int cap, int cost) {
20         Edge e1 = {v, cost, cap, 0, (int) g[v].size()};
21         Edge e2 = {u, -cost, 0, 0, (int) g[u].size()};
22         g[u].push_back(e1);
23         g[v].push_back(e2);
24     }
25
26     pair<int, int> minCostMaxFlow(int s, int t) {
27         int flow = 0;
28         int cost = 0;
29         vector<int> state(n), from(n), from_edge(n);
30         vector<int> d(n);
31         deque<int> q;
32         while (true) {
33             for (int i = 0; i < n; i++)
34                 state[i] = 2, d[i] = inf, from[i] = -1;
35             state[s] = 1;
36             q.clear();
37             q.push_back(s);
38             d[s] = 0;
39             while (!q.empty()) {
40                 int v = q.front();
41                 q.pop_front();
42                 state[v] = 0;
43                 for (int i = 0; i < (int) g[v].size(); i++) {
44                     Edge e = g[v][i];
45                     if (e.flow >= e.cap || (d[e.to] <= d[v] + e.cost))
46                         continue;
47                     int to = e.to;
48                     d[to] = d[v] + e.cost;
49                     from[to] = v;
50                     from_edge[to] = i;
51                     if (state[to] == 1) continue;
52                     if (!state[to] || (!q.empty() && d[q.front()] > d[
53                         to]))
54                         q.push_front(to);
55                     else q.push_back(to);
56                     state[to] = 1;
57                 }
58             }
59             if (d[t] == inf) break;
60             int it = t, addflow = inf;
61             while (it != s) {
62                 addflow = min(addflow,

```

```

62                 g[from[it]][from_edge[it]].cap
63                 - g[from[it]][from_edge[it]].flow);
64                 it = from[it];
65             }
66             it = t;
67             while (it != s) {
68                 g[from[it]][from_edge[it]].flow += addflow;
69                 g[it][g[from[it]][from_edge[it]].backEdge].flow -=
50                     addflow;
70                 cost += g[from[it]][from_edge[it]].cost * addflow;
71                 it = from[it];
72             }
73             flow += addflow;
74         }
75         return {cost, flow};
76     }
77 };

```

3.7 Matching

3.7.1 HopcroftKarp

```

1 struct HopcroftKarp {
2     vector<int> leftMatch, rightMatch, dist, cur;
3     vector<vector<int>> > a;
4     int n, m;
5
6     HopcroftKarp() {}
7
8     HopcroftKarp(int n, int m) {
9         this->n = n;
10        this->m = m;
11        a = vector<vector<int>> >(n);
12        leftMatch = vector<int>(m, -1);
13        rightMatch = vector<int>(n, -1);
14        dist = vector<int>(n, -1);
15        cur = vector<int>(n, -1);
16    }
17
18    void addEdge(int x, int y) {
19        a[x].push_back(y);
20    }
21
22    int bfs() {
23        int found = 0;
24        queue<int> q;
25        for (int i = 0; i < n; i++)
26            if (rightMatch[i] < 0) dist[i] = 0, q.push(i);
27        else dist[i] = -1;
28
29        while (!q.empty()) {
30            int x = q.front();
31            q.pop();
32            for (int i = 0; i < int(a[x].size()); i++) {
33                int y = a[x][i];
34                if (leftMatch[y] < 0) found = 1;
35                else if (dist[leftMatch[y]] < 0)
36                    dist[leftMatch[y]] = dist[x] + 1, q.push(leftMatch[
37                        y]);
38            }
39        }
40        return found;
41    }
42
43    int dfs(int x) {
44        for (; cur[x] < int(a[x].size()); cur[x]++) {
45            int y = a[x][cur[x]];
46            if (leftMatch[y] < 0 || (dist[leftMatch[y]] == dist[x] + 1
47                && dfs(leftMatch[y]))) {
48                leftMatch[y] = x;
49                rightMatch[x] = y;
50                return 1;
51            }
52        }
53        return 0;
54    }
55
56    int maxMatching() {
57        int match = 0;
58        while (bfs()) {
59            for (int i = 0; i < n; i++) cur[i] = 0;
60            for (int i = 0; i < n; i++)
61                if (rightMatch[i] < 0) match += dfs(i);
62        }
63        return match;
64    };

```

4 Number Theory

4.1 Euclidean Algorithm

4.1.1 GCD_LCM

```

1 ll gcd(ll a, ll b) {
2     if (b == 0)
3         return a;
4     return gcd(b, a % b);
5     // return __gcd(a, b);
6 }
7
8 ll lcm(ll a, ll b) {
9     return (a / gcd(a, b)) * b;
10 }

```

4.1.2 ExtendedEuclid

```

1 long long extGCD(long long a, long long b, long long &x, long long &y)
2 {
3     if (b == 0) {
4         x = 1; y = 0;
5         return a;
6     }

```

```

5     }
6     long long x1, y1;
7     long long d = extGCD(b, a % b, x1, y1);
8     x = y1;
9     y = x1 - y1 * (a / b);
10    return d;
11 }
12
13 long long modInverse(long long a, long long m) {
14     long long x, y;
15     long long g = extGCD(a, m, x, y);
16     if (g != 1) return -1; // Inverse doesn't exist
17     return (x % m + m) % m;
18 }

```

4.2 Primes and Divisors

4.2.1 Sieve

```

1 bool isPrime[N];
2
3 void seive(int n = N) {
4     //complexity: n*log(log(n))
5     memset(isPrime, true, sizeof isPrime);
6     isPrime[0] = false;
7     isPrime[1] = false;
8     for (int i = 2; i * i < n; i++) {
9         if (isPrime[i]) {
10            for (int j = i * i; j < n; j += i) {
11                isPrime[j] = false;
12            }
13        }
14    }
15 }

```

4.2.2 SPF

```

1 vector<int> spf;
2
3 void seive(int n = N) {
4     //complexity: n*log(log(n))
5     spf.resize(n);
6     iota(all(spf), 0);
7     for (int i = 2; i * i < n; i++) {
8         if (spf[i] == i) {
9             for (int j = i * i; j < n; j += i) {
10                spf[j] = min(spf[j], i);
11            }
12        }
13    }
14 }
15
16 vector<int> getPrimeFactors(int n) {
17     //complexity: log(n)
18     vector<int> ans;
19     while (n != 1) {
20         ans.pb(spf[n]);
21         n /= spf[n];
22     }
23     return ans;
24 }

```

4.2.3 Divisors

```

1 template <typename T>
2 vector<T> getDivisors(T n) {
3     //complexity: sqrt(n)
4     vector<T> vec;
5     for (ll i = 1; i * i <= n; i++) {
6         if (n % i == 0) {
7             vec.push_back(i);
8             if (n / i != i)
9                 vec.push_back(n / i);
10        }
11    }
12    return vec;
13 }

```

4.2.4 Factorization

```

1 template <typename T>
2 vector<T> getPrimeFactors(T n) {
3     //complexity: sqrt(n)
4     vector<T> vec;
5     for (ll i = 2; i * i <= n; i++) {
6         while (n % i == 0) {
7             vec.push_back(i);
8             n /= i;
9         }
10    }
11    if (n != 1) vec.push_back(n);
12    return vec;
13 }

```

4.3 Modular Arithmetic

4.3.1 CRT

```

1 long long extGCD(long long a, long long b, long long &x, long long &y)
2 {
3     if (b == 0) {
4         x = 1; y = 0;
5         return a;
6     }
7     long long x1, y1;
8     long long d = extGCD(b, a % b, x1, y1);
9     x = y1;
10    y = x1 - y1 * (a / b);
11    return d;
12 }
13 // Solves x = a_i (mod m_i). Returns {x, lcm(m_1, ..., m_k)}.

```

```

14 pair<long long, long long> CRT(const vector<long long>& a, const vector
15     <long long>& m) {
16     long long res = a[0], lcm = m[0];
17     for (int i = 1; i < a.size(); i++) {
18         long long x, y;
19         long long g = extGCD(lcm, m[i], x, y);
20         if ((a[i] - res) % g != 0) return {-1, -1}; // No solution
21
22         long long step = m[i] / g;
23         long long k = ((a[i] - res) / g) % step * (x % step) % step;
24         res += k * lcm;
25         lcm *= step;
26         res = (res % lcm + lcm) % lcm;
27     }
28     return {res, lcm};
29 }

```

5 Combinatorics

6 Math

6.1 Modular Arithmetic

6.1.1 ModularArithmetic

```

1 int fact[N], inv[N], invFact[N]; //used in nCr(), nPr(), calcFact()
2
3 int modSum(ll a, ll b) {
4     if (a < 0)
5         a += MOD;
6     if (b < 0)
7         b += MOD;
8     a += b;
9     if (a >= MOD)
10        a -= MOD;
11    return a;
12
13    a = (a % MOD + MOD) % MOD;
14    b = (b % MOD + MOD) % MOD;
15    return (a + b) % MOD;
16 }
17
18 int modProd(ll a, ll b) {
19     if (a < 0)
20         a += MOD;
21     if (b < 0)
22         b += MOD;
23     a *= b;
24     if (a >= MOD)
25         a %= MOD;
26    return a;
27
28    a = (a % MOD + MOD) % MOD;
29    b = (b % MOD + MOD) % MOD;
30    return (a * b) % MOD;
31 }
32
33 int modPow(int a, ll b) {
34     int result = 1;
35     while (b) {
36         if (b & 1)
37             result = modProd(result, a);
38         a = modProd(a, a);
39         b >>= 1;
40     }
41     return result;
42 }
43
44 int modInverse(int a) {
45     return modPow(a, MOD - 2);
46 }
47
48 int modDiv(int a, int b) {
49     return modProd(a, modInverse(b));
50 }
51
52 int nCr(int n, int r) {
53     if (r < 0 || r > n)
54         return 0;
55     // return modDiv(fact[n], modProd(fact[r], fact[n - r]));
56     return modProd(fact[n], modProd(invFact[r], invFact[n - r]));
57 }
58
59 int nPr(int n, int r) {
60     if (r < 0 || r > n)
61         return 0;
62     // return modDiv(fact[n], fact[n - r]);
63     return modProd(fact[n], invFact[n - r]);
64 }
65
66 void build() {
67     fact[0] = invFact[0] = 1;
68     for (int i = 1; i < N; i++) {
69         fact[i] = modProd(fact[i - 1], i);
70         invFact[i] = modDiv(invFact[i - 1], i);
71         inv[i] = modInverse(i);
72     }
73 }
74
75 }
76 }

```

6.1.2 ModularIntClass

```

1 struct Mint {
2     int v;
3
4     Mint(long long val = 0) {
5         v = int(val % MOD);
6         if (v < 0) v += MOD;
7     }
8
9     Mint operator+(const Mint& o) const { return Mint(v + o.v); }

```

```

10 Mint operator-(const Mint& o) const { return Mint(v - o.v); }
11 Mint operator*(const Mint& o) const { return Mint(1LL * v * o.v); }
12 Mint operator/(const Mint& o) const { return *this * o.inv(); }
13
14 Mint& operator+=(const Mint& o) {
15     v += o.v;
16     if (v >= MOD) v -= MOD;
17     return *this;
18 }
19
20 Mint& operator-=(const Mint& o) {
21     v -= o.v;
22     if (v < 0) v += MOD;
23     return *this;
24 }
25
26 Mint& operator*=(const Mint& o) {
27     v = int(1LL * v * o.v % MOD);
28     return *this;
29 }
30
31 Mint& operator/=(const Mint& o) {
32     v = int(1LL * v * o.inv().v % MOD);
33     return *this;
34 }
35
36 Mint pow(long long p) const {
37     Mint a = *this, res = 1;
38     while (p > 0) {
39         if (p & 1) res *= a;
40         a *= a;
41         p >>= 1;
42     }
43     return res;
44 }
45
46 Mint inv() const { return pow(MOD - 2); }
47
48 friend ostream& operator<<(ostream& os, const Mint& m) {
49     os << m.v;
50     return os;
51 }
52 };
53
54 Mint fact[N], inv[N], invFact[N];
55
56 Mint nCr(int n, int r) {
57     if (r < 0 || r > n)
58         return 0;
59     // return fact[n] / (fact[r] * fact[n - r]);
60     return fact[n] * invFact[r] * invFact[n - r];
61 }
62
63 Mint nPr(int n, int r) {
64     if (r < 0 || r > n)
65         return 0;
66     // return fact[n] / fact[n - r];
67     return fact[n] * invFact[n - r];
68 }
69
70 void build() {
71     fact[0] = invFact[0] = 1;
72     for (int i = 1; i < N; i++) {
73         fact[i] = fact[i - 1] * i;
74         invFact[i] = invFact[i - 1] / i;
75         inv[i] = Mint(i).inv();
76     }
77 }

```

6.2 Multiplicative Functions

6.2.1 EulerTotientPhi

```

1 int phi[N];
2
3 void calculatePhi() {
4     for (int i = 0; i < N; ++i) phi[i] = i & 1 ? i : i / 2;
5     for (int i = 3; i < N; i += 2)
6         if (phi[i] == i)
7             for (int j = i; j < N; j += i) phi[j] -= phi[j] / i;
8 }

```

6.3 Matrices

6.3.1 MatrixExponentiation

```

1 int modSum(ll a, ll b) {
2     if (a < 0)
3         a += MOD;
4     if (b < 0)
5         b += MOD;
6     a += b;
7     if (a >= MOD)
8         a -= MOD;
9     return a;
10 }
11
12 int modProd(ll a, ll b) {
13     if (a < 0)
14         a += MOD;
15     if (b < 0)
16         b += MOD;
17     a *= b;
18     if (a >= MOD)
19         a %= MOD;
20     return a;
21 }
22
23 typedef vector<vector<int>> matrix;
24
25 matrix operator*(const matrix& lhs, const matrix& rhs) {
26     int n = lhs.size();
27     int m = rhs[0].size();
28     int s1 = lhs[0].size(), s2 = rhs.size();

```

```

29     assert(s1 == s2);
30     matrix ret(n, vector<int>(m));
31     for (int i = 0; i < n; ++i)
32         for (int j = 0; j < s1; ++j)
33             for (int k = 0; k < m; ++k)
34                 ret[i][k] = modSum(ret[i][k], modProd(lhs[i][j], rhs[j][k]));
35     return ret;
36 }
37
38 matrix Identity(int n) {
39     matrix ret(n, vector<int>(n));
40     for (int i = 0; i < n; ++i) {
41         ret[i][i] = 1;
42     }
43     return ret;
44 }
45
46 matrix mat_power(matrix x, ll p) {
47     matrix res = Identity(x.size());
48     while (p) {
49         if (p & 1) res = (res * x);
50         x = (x * x);
51         p >>= 1;
52     }
53     return res;
54 }

```

6.3.2 FastMatrixPower

```

1 const int SZ = 2, mod = 1e9 + 7;
2
3 int modSum(ll a, ll b) {
4     if (a < 0)
5         a += mod;
6     if (b < 0)
7         b += mod;
8     a += b;
9     if (a >= mod)
10        a -= mod;
11    return a;
12 }
13
14 int modProd(ll a, ll b) {
15     if (a < 0)
16         a += mod;
17     if (b < 0)
18         b += mod;
19     a *= b;
20     if (a >= mod)
21         a %= mod;
22     return a;
23 }
24
25 typedef array<array<int, SZ>, SZ> matrix;
26
27 matrix operator*(const matrix &lhs, const matrix &rhs) {
28     matrix ret{};
29     for (int i = 0; i < SZ; ++i)
30         for (int j = 0; j < SZ; ++j)
31             for (int k = 0; k < SZ; ++k)
32                 ret[i][k] = modSum(ret[i][k], modProd(lhs[i][j], rhs[j][k]));
33     return ret;
34 }
35
36 matrix Identity(int n) {
37     matrix ret = {};
38     for (int i = 0; i < n; ++i) {
39         ret[i][i] = 1;
40     }
41     return ret;
42 }
43
44 matrix mat_power(matrix x, ll p) {
45     matrix res = Identity(x.size());
46     while (p) {
47         if (p & 1) res = (res * x);
48         x = (x * x);
49         p >>= 1;
50     }
51     return res;
52 }

```

6.4 Linear Algebra

6.4.1 LinearXorBasis

```

1
2
3 // Linear XOR Basis - Vector Space over GF(2)
4 // Time: O(LOG_BITS) per insert/query
5 struct Basis {
6     vector<ll> basis;
7     int log_bits, sz;
8
9     Basis(int log_bits = 60) : log_bits(log_bits), sz(0), basis(
10         log_bits, 0) {}
11
12     bool insert(ll x) {
13         for (int i = log_bits - 1; i >= 0; i--) {
14             if (!(x & (1LL << i))) continue;
15             if (!basis[i]) {
16                 basis[i] = x;
17                 sz++;
18                 return true;
19             }
20             x ^= basis[i];
21         }
22         return false;
23     }
24
25     ll getMax(ll res = 0) const {
26         for (int i = log_bits - 1; i >= 0; i--) {

```

```

26         res = max(res, res ^ basis[i]);
27     }
28     return res;
29 }
30 };

```

6.5 Convolution

6.5.1 FFT

```

1 #define rep(aa, bb, cc) for(int aa = bb; aa < cc; aa++)
2 #define sz(a) (int)a.size()
3 typedef complex<double> C;
4 typedef vector<double> vd;
5 void fft(vector<C>& a) {
6     int n = sz(a), L = 31 - __builtin_clz(n);
7     static vector<complex<long double>> R(2, 1);
8     static vector<C> rt(2, 1); // (~10% faster if double)
9     for (static int k = 2; k < n; k *= 2) {
10         R.resize(n); rt.resize(n);
11         auto x = polar(1.0L, acos(-1.0L) / k);
12         rep(i, k, 2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
13     }
14     vi rev(n);
15     rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
16     rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
17     for (int k = 1; k < n; k *= 2)
18         for (int i = 0; i < n; i += 2 * k) rep(j, 0, k) {
19             // C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-
20             // rolled) // include-line
21             auto x = (double *)&rt[j+k], y = (double *)&a[i+j+k];
22             // exclude-line
23             C z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
24             // exclude-line
25             a[i + j + k] = a[i + j] - z;
26             a[i + j] += z;
27 }
28 template<int M> vi convMod(const vi &a, const vi &b) {
29     if (a.empty() || b.empty()) return {};
30     vi res(sz(a) + sz(b) - 1);
31     int B=32-__builtin_clz(sz(res)), n=1<B, cut=int(sqrt(M));
32     vector<C> L(n), R(n), outs(n), outl(n);
33     rep(i, 0, sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
34     rep(i, 0, sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
35     fft(L), fft(R);
36     rep(i, 0, n) {
37         int j = -i & (n - 1);
38         outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
39         outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / ii;
40     }
41     fft(outl), fft(outs);
42     rep(i, 0, sz(res)) {
43         ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])+.5);
44         ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
45         res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
46     }
47     return res;
48 }

```

6.5.2 NTT

```

1 const int MOD = 998244353;
2 const int ROOT = 3; // Primitive root for 998244353
3
4 long long power(long long base, long long exp) {
5     long long res = 1;
6     base %= MOD;
7     while (exp > 0) {
8         if (exp % 2 == 1) res = (res * base) % MOD;
9         base = (base * base) % MOD;
10        exp /= 2;
11    }
12    return res;
13 }
14
15 long long modInverseFast(long long n) {
16     return power(n, MOD - 2);
17 }
18
19 void ntt(vector<long long>& a, bool invert) {
20     int n = a.size();
21     for (int i = 1, j = 0; i < n; i++) {
22         int bit = n >> 1;
23         for (; j & bit; bit >>= 1) j ^= bit;
24         j ^= bit;
25         if (i < j) swap(a[i], a[j]);
26     }
27     for (int len = 2; len <= n; len <= 1) {
28         long long wlen = power(ROOT, (MOD - 1) / len);
29         if (invert) wlen = modInverseFast(wlen);
30         for (int i = 0; i < n; i += len) {
31             long long w = 1;
32             for (int j = 0; j < len / 2; j++) {
33                 long long u = a[i + j], v = (a[i + j + len / 2] * w) % MOD;
34                 a[i + j] = u + v < MOD ? u + v : u + v - MOD;
35                 a[i + j + len / 2] = u - v >= 0 ? u - v : u - v + MOD;
36                 w = (w * wlen) % MOD;
37             }
38         }
39     }
40     if (invert) {
41         long long n_inv = modInverseFast(n);
42         for (long long& x : a) x = (x * n_inv) % MOD;
43     }
44 }

```

7 Strings

7.1 String Matching

7.1.1 KMP

```

1 vector<int> compute_pi(const string &s) {
2     int n = s.size();
3     vector<int> pi(n);
4     for (int i = 1; i < n; i++) {
5         int j = pi[i - 1];
6         while (j > 0 && s[i] != s[j])
7             j = pi[j - 1];
8         if (s[i] == s[j])
9             j++;
10        pi[i] = j;
11    }
12    return pi;
13 }
14
15 vector<int> find_matches(const string &text, const string &pat) {
16     int n = pat.length(), m = text.length();
17     string s = pat + "#" + text;
18     vector<int> pi = compute_pi(s), ans;
19     for (int i = n + 1; i <= n + m; i++) { /* n + 1 is where the text
20         starts */
21         if (pi[i] == n) {
22             ans.push_back(i - 2 * n); /* i - (n - 1) - (n + 1) */
23         }
24     }
25     return ans;
26 }
27
28 void compute_automaton(string s, vector<vector<int>> &aut) {
29     s += '#';
30     int n = s.size();
31     vector<int> pi = compute_pi(s);
32     aut.assign(n, vector<int>(26));
33     for (int i = 0; i < n; i++) {
34         for (int c = 0; c < 26; c++) {
35             if (i > 0 && 'a' + c != s[i])
36                 aut[i][c] = aut[pi[i - 1]][c];
37             else
38                 aut[i][c] = i + ('a' + c == s[i]);
39         }
40     }
41 }

```

7.1.2 RabinKarp

```

1 vector<int> rabin_karp(string const &s, string const &t) {
2     const int p = 31;
3     const int m = 1e9 + 9;
4     int S = s.size(), T = t.size();
5
6     vector<ll> p_pow(max(S, T));
7     p_pow[0] = 1;
8     for (int i = 1; i < (int) p_pow.size(); i++)
9         p_pow[i] = (p_pow[i - 1] * p) % m;
10
11     vector<ll> h(T + 1, 0);
12     for (int i = 0; i < T; i++)
13         h[i + 1] = (h[i] + (t[i] - 'a' + 1) * p_pow[i]) % m;
14     ll h_s = 0;
15     for (int i = 0; i < S; i++)
16         h_s = (h_s + (s[i] - 'a' + 1) * p_pow[i]) % m;
17
18     vector<int> occurrences;
19     for (int i = 0; i + S - 1 < T; i++) {
20         ll cur_h = (h[i + S] + m - h[i]) % m;
21         if (cur_h == h_s * p_pow[i] % m)
22             occurrences.push_back(i);
23     }
24     return occurrences;
25 }

```

7.1.3 ZFunction

```

1 // z[i] is the length of the longest substring starting from s[i] which
2 // is also a prefix of s
3 vector<int> z_function(string s) {
4     int n = s.length();
5     vector<int> z(n);
6     int l = 0, r = 0;
7     for (int i = 1; i < n; i++) {
8         if (i < r) {
9             z[i] = min(r - i, z[i - l]);
10        }
11        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
12            z[i]++;
13        }
14        if (i + z[i] > r) {
15            l = i;
16            r = i + z[i];
17        }
18    }
19    return z;
20 }

```

7.2 Hashing

7.2.1 StringHashing

```

1 int modSum(ll a, ll b) {
2     //complexity: 1
3     if (a < 0)
4         a += MOD;
5     if (b < 0)
6         b += MOD;
7     a += b;
8     if (a >= MOD)
9         a %= MOD;
10    return a;
11 }

```

```

12
13 int modProd(ll a, ll b) {
14     ///complexity: 1
15     if (a < 0)
16         a += MOD;
17     if (b < 0)
18         b += MOD;
19     a *= b;
20     if (a >= MOD)
21         a %= MOD;
22     return a;
23 }
24
25 int fastPow(int a, ll b) {
26     ///complexity: log(b)
27     if (b == 0) return 1;
28     int temp = fastPow(a, b / 2);
29     if (b % 2 == 0)
30         return modProd(temp, temp);
31     return modProd(modProd(a, temp), temp);
32 }
33
34
35 int modInverse(int a) {
36     ///complexity: log(mod)
37     return fastPow(a, MOD - 2);
38 }
39
40
41 struct Hash {
42     vector<pair<int, int>> hash_value = {{0, 0}};
43     const pair<int, int> p = {67, 97};
44     vector<pair<int, int>> p_pow = {{1, 1}};
45     vector<pair<int, int>> inv = {{1, 1}};
46
47     void compute(string const &s) {
48         for (char c: s) {
49             if (c >= 'a' && c <= 'z') {
50                 c = c - 'a' + 1;
51             } else if (c >= 'A' && c <= 'Z') {
52                 c = c - 'A' + 30;
53             } else {
54                 c = c - '0' + 60;
55             }
56             int x = modSum(hash_value.back().st, modProd(c, p_pow.back().st));
57             int a = modProd(p_pow.back().st, p.st);
58
59             int y = modSum(hash_value.back().nd, modProd(c, p_pow.back().nd));
60             int b = modProd(p_pow.back().nd, p.nd);
61
62             hash_value.emplace_back(x, y);
63             p_pow.emplace_back(a, b);
64             if (inv.size() == 1) {
65                 inv.emplace_back(modInverse(a), modInverse(b));
66             } else {
67                 inv.emplace_back(modProd(inv[1].st, inv.back().st),
68                                     modProd(inv[1].nd, inv.back().nd));
69             }
70         }
71     }
72
73     pair<int, int> get(int l, int r) {
74         l++, r++;
75         pair<int, int> ans;
76         ans.st = modSum(hash_value[r].st, -hash_value[l - 1].st);
77         ans.st = modProd(ans.st, inv[l].st);
78
79         ans.nd = modSum(hash_value[r].nd, -hash_value[l - 1].nd);
80         ans.nd = modProd(ans.nd, inv[l].nd);
81
82         return ans;
83     }
84 };

```

7.3 Aho Corasick

7.3.1 AhoCorasick

```

1 struct Mapper {
2     int operator()(char c) const { return c - 'a'; }
3 };
4
5 template <int ALPHABET, typename Mapper>
6 struct Aho {
7     struct Node {
8         array<int, ALPHABET> nxt;
9         int link = 0;
10        int terminal_idx = -1; /// -1 if not a terminal, else idx of
11            string insertion
12        int exit_link = 0; /// Points to the nearest terminal suffix
13        int depth = 0; /// Size of the prefix the node represents
14        Node() { nxt.fill(0); }
15    };
16
17    vector<Node> t;
18    vector<int> bfs_order;
19    Mapper get_idx; /// Instance of the mapping function
20    int word_count = 0; /// Tracks the order of inserted strings
21
22    /// Constructor optionally accepts a custom mapper instance
23    Aho(Mapper mapper = Mapper()) : get_idx(mapper) {
24        t.emplace_back();
25    }
26
27    int insert(const string& p) {
28        int u = 0;
29        for (char c : p) {
30            int v = get_idx(c);
31            if (!t[u].nxt[v]) {
32                t[u].nxt[v] = t.size();
33                t.emplace_back();

```

```

33        t.back().depth = t[u].depth + 1; /// New node's depth is
34            parent's depth + 1
35    }
36    u = t[u].nxt[v];
37
38    /// If this exact string hasn't been inserted before, assign it
39    the current index
40    if (t[u].terminal_idx == -1) {
41        t[u].terminal_idx = word_count;
42    }
43    word_count++; /// Increment for the next insertion
44
45    return u;
46 }
47
48 void build() {
49     queue<int> q;
50     for (int c = 0; c < ALPHABET; ++c) {
51         if (t[0].nxt[c]) {
52             q.push(t[0].nxt[c]);
53         }
54     }
55
56     while (!q.empty()) {
57         int u = q.front();
58         q.pop();
59         bfs_order.push_back(u);
60
61         for (int c = 0; c < ALPHABET; ++c) {
62             int v = t[u].nxt[c];
63             if (v) {
64                 t[v].link = t[t[u].link].nxt[c];
65
66                 /// Compute the exit link:
67                 /// If the immediate suffix link is a terminal,
68                 point to it.
69                 /// Otherwise, copy its exit link.
70                 t[v].exit_link = (t[t[v].link].terminal_idx != -1)
71                     ? t[v].link : t[t[v].link].exit_link;
72             }
73             q.push(v);
74         }
75     }
76 }
77
78 int advance(int u, char c) {
79     return t[u].nxt[get_idx(c)];
80 }
81
82 };

```

7.4 Suffix Structures

7.4.1 SuffixArray

```

1 struct SuffixArray {
2     string S;
3     /// sa is the suffix array with the empty suffix being sa[0]
4     /// lcp[i] holds the lcp between sa[i], sa[i - 1]
5     vector<int> logs, sa, lcp, rank;
6     vector<vector<int>> table;
7
8     SuffixArray() {};
9
10    SuffixArray(string &s, int lim = 256) {
11        S = s;
12        int n = s.size() + 1, k = 0, a, b;
13        vector<int> c(s.begin(), s.end() + 1), tmp(n), frq(max(n, lim))
14            ;
15        c.back() = 0; /// 0 is less than any character
16        sa = lcp = rank = tmp, iota(sa.begin(), sa.end(), 0);
17        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
18            p = j, iota(tmp.begin(), tmp.end(), n - j);
19            for (int i = 0; i < n; i++) {
20                if (sa[i] >= j)
21                    tmp[p++] = sa[i] - j;
22            }
23
24            fill(frq.begin(), frq.end(), 0);
25
26            for (int i = 0; i < n; i++) frq[c[i]]++;
27            for (int i = 1; i < lim; i++) frq[i] += frq[i - 1];
28            for (int i = n; i--;) sa[--frq[c[tmp[i]]]] = tmp[i];
29
30            swap(c, tmp), p = 1, c[sa[0]] = 0;
31            for (int i = 1; i < n; i++)
32                a = sa[i - 1], b = sa[i], c[b] = (tmp[a] == tmp[b] &&
33                    tmp[a + j] == tmp[b + j]) ? p - 1 : p++;
34        }
35
36        for (int i = 1; i < n; i++) rank[sa[i]] = i;
37        for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
38            for (k &&k--, j = sa[rank[i] - 1];
39                 s[i + k] == s[j + k]; k++);
40    }
41
42    void preLcp() {
43        int n = S.size() + 1;
44        logs = vector<int>(n + 5);
45        for (int i = 2; i < n + 5; ++i) {
46            logs[i] = logs[i / 2] + 1;
47        }
48        table = vector<vector<int>>(n, vector<int>(20));
49        for (int i = 0; i < n; ++i) {
50            table[i][0] = lcp[i];
51        }
52
53        for (int j = 1; j <= logs[n]; ++j) {
54            for (int i = 0; i <= n - (1 << j); ++i) {

```

```

54         table[i][j] = min(table[i][j] - 1], table[i + (1 << (j -
55             1))][j - 1]);
56     }
57 }
58
59 int queryLcp(int i, int j) {
60     // if (i == j) return (int) S.size() - i;
61     // i = rank[i], j = rank[j];
62     if (i == j) return (int) S.size() - sa[i];
63     if (i > j)
64         swap(i, j);
65     i++;
66     int len = logs[j - i + 1];
67     return min(table[i][len], table[j - (1 << len) + 1][len]);
68 }
69 };

```

7.4.2 SuffixAutomaton

```

1 struct SuffixAutomaton {
2     struct Node {
3         int len = 0, link = -1;
4         map<char, int> next;
5     };
6
7     vector<Node> st;
8     int sz = 1, last = 0;
9
10    SuffixAutomaton(const string& s) {
11        st.resize(s.length() * 2);
12        st[0].len = 0;
13        st[0].link = -1;
14        for (char c : s) extend(c);
15    }
16
17    void extend(char c) {
18        int cur = sz++;
19        st[cur].len = st[last].len + 1;
20        int p = last;
21        while (p != -1 && !st[p].next.count(c)) {
22            st[p].next[c] = cur;
23            p = st[p].link;
24        }
25        if (p == -1) {
26            st[cur].link = 0;
27        } else {
28            int q = st[p].next[c];
29            if (st[p].len + 1 == st[q].len) {
30                st[cur].link = q;
31            } else {
32                int clone = sz++;
33                st[clone].len = st[p].len + 1;
34                st[clone].next = st[q].next;
35                st[clone].link = st[q].link;
36                while (p != -1 && st[p].next[c] == q) {
37                    st[p].next[c] = clone;
38                    p = st[p].link;
39                }
40                st[q].link = st[cur].link = clone;
41            }
42        }
43        last = cur;
44    }
45 };

```

7.5 Palindromes

7.5.1 Manacher

```

1 struct Manacher {
2     vector<int> p;
3
4     // Returns a vector where p[i] is the radius of the longest
5     // Centers are interleaved: 0 is before string, 1 is first char, 2
6     // is between 1st & 2nd, etc.
7     Manacher(string s) {
8         string t = "#";
9         for (char c : s) {
10             t += c;
11             t += "#";
12         }
13         int n = t.size();
14         p.assign(n, 0);
15         int c = 0, r = 0;
16         for (int i = 0; i < n; i++) {
17             int mirror = 2 * c - i;
18             if (i < r) p[i] = min(r - i, p[mirror]);
19             while (i - 1 - p[i] >= 0 && i + 1 + p[i] < n && t[i - 1 - p[i]] == t[i + 1 + p[i]]) {
20                 p[i]++;
21             }
22             if (i + p[i] > r) {
23                 c = i;
24                 r = i + p[i];
25             }
26         }
27
28         // Check if substring [l, r] (0-based) is a palindrome in O(1)
29         bool is_palindrome(int l, int r) {
30             int len = r - l + 1;
31             int center = l + r + 1;
32             return p[center] >= len;
33         }
34 };

```

8 Dynamic Programming (DP)

8.1 DP Optimizations

8.1.1 DivideAndConquerDP

```

1 int a[N];
2 ll curCost;
3 int curL = 0, curR = -1, freq[N];
4 pair<int, ll> pre[N], cur[N];
5 // Modify at will :
6 int lo(int ind) { return 1; }
7 int hi(int ind) { return ind; }
8
9 void add(int val) {
10     curCost += freq[val]++;
11 }
12
13 void remove(int val) {
14     curCost -= --freq[val];
15 }
16
17 ll Cost(int l, int r) {
18     while (curR < r)
19         add(a[++curR]);
20     while (curL > l)
21         add(a[--curL]);
22     while (curR > r)
23         remove(a[curR--]);
24     while (curL < l)
25         remove(a[curL++]);
26     return curCost;
27 }
28
29 ll f(int ind, int k) { return pre[k - 1].nd + Cost(k, ind); }
30 void store(int ind, int k, ll v) { cur[ind] = pair(k, v); }
31
32 void rec(int L, int R, int LO, int HI) {
33     if (L > R) return;
34     int mid = (L + R) >> 1;
35     pair best(llinf, LO);
36     for (int k = max(LO, lo(mid)); k <= min(HI, hi(mid)); ++k)
37         best = min(best, pair(f(mid, k), k));
38     store(mid, best.second, best.first);
39     rec(L, mid - 1, LO, best.second);
40     rec(mid + 1, R, best.second, HI);
41 }
42
43 void solve(int L, int R) { rec(L, R, -inf, inf); }

```

8.1.2 CHT

```

1 //To query the maximum value of the function for all x in [l, r], query
2 // only two points: l, r because the functions are convex.
3 struct Line {
4     ll m, b;
5     mutable function<const Line *(> succ;
6
7     bool operator<(const Line &other) const {
8         return m < other.m;
9     }
10
11     bool operator<(const ll &x) const {
12         const Line *s = succ();
13         if (!s)
14             return 0;
15         return b - s->b < (s->m - m) * x;
16     }
17 };
18
19 // will maintain upper hull for maximum
20 struct HullDynamic : public multiset<Line, less<>> {
21     bool bad(iterator y) {
22         auto z = next(y);
23         if (y == begin()) {
24             if (z == end())
25                 return 0;
26             return y->m == z->m && y->b <= z->b;
27         }
28         auto x = prev(y);
29         if (z == end())
30             return y->m == x->m && y->b <= x->b;
31         return (ld)(x->b - y->b) * (z->m - y->m) >= (ld)(y->b - z->b) * (y->m - x->m);
32     }
33
34     void insert_line(ll m, ll b) {
35         // for minimum
36         m *= -1;
37         b *= -1;
38         auto y = insert({m, b});
39         y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };
40         if (bad(y)) {
41             erase(y);
42             return;
43         }
44         while (next(y) != end() && bad(next(y)))
45             erase(next(y));
46         while (y != begin() && bad(prev(y)))
47             erase(prev(y));
48     }
49
50     ll query(ll x) {
51         auto l = *lower_bound(x);
52         // for minimum
53         return -(l.m * x + l.b);
54     }
55 };

```

8.1.3 CHTRollback

```

1 // Template parameters:
2 // IS_MAX: Set to true for Maximum queries, false for Minimum queries.

```

```

3 // IS_INC: Set to true if slopes added are strictly Increasing, false
  for Decreasing.
4 template<bool IS_MAX = false, bool IS_INC = false>
5 struct RollbackCHT {
6     struct Line {
7         ll m, c;
8         Line() : m(0), c(llinf) {}
9         Line(ll m, ll c) : m(m), c(c) {}
10        ll eval(ll x) const { return m * x + c; }
11    };
12
13    struct History {
14        int pos;
15        int sz;
16        Line old_line;
17    };
18
19    vector<Line> st;
20    vector<History> hist;
21    int sz;
22
23    RollbackCHT(int max_nodes) {
24        st.resize(max_nodes);
25        sz = 0;
26    }
27
28    bool redundant(Line l1, Line l2, Line l3) {
29        __int128 dx1 = l1.m - l2.m;
30        __int128 dc1 = l2.c - l1.c;
31        __int128 dx2 = l2.m - l3.m;
32        __int128 dc2 = l3.c - l2.c;
33        return dc1 * dx2 >= dc2 * dx1;
34    }
35
36    void add(ll m, ll c) {
37        // The magic happens here: automatically formats internal state
38        ll m_int = IS_INC ? -m : m;
39        ll c_int = IS_MAX ? -c : c;
40
41        Line l_new(m_int, c_int);
42        int pos = sz;
43        int low = 1, high = sz - 1;
44
45        while (low <= high) {
46            int mid = low + (high - low) / 2;
47            if (redundant(st[mid - 1], st[mid], l_new)) {
48                pos = mid;
49                high = mid - 1;
50            } else {
51                low = mid + 1;
52            }
53        }
54
55        hist.push_back({pos, sz, st[pos]});
56        st[pos] = l_new;
57        sz = pos + 1;
58    }
59
60    void rollback() {
61        if (hist.empty()) return;
62        History h = hist.back();
63        hist.pop_back();
64        sz = h.sz;
65        st[h.pos] = h.old_line;
66    }
67
68    ll query(ll x) {
69        if (sz == 0) return IS_MAX ? -llinf : llinf;
70
71        // The magic happens here: matches x to the internal geometry
72        ll x_int = (IS_INC != IS_MAX) ? -x : x;
73
74        int low = 0, high = sz - 2;
75        int best = sz - 1;
76
77        while (low <= high) {
78            int mid = low + (high - low) / 2;
79            if (st[mid].eval(x_int) <= st[mid + 1].eval(x_int)) {
80                best = mid;
81                high = mid - 1;
82            } else {
83                low = mid + 1;
84            }
85        }
86
87        ll ans = st[best].eval(x_int);
88        return IS_MAX ? -ans : ans;
89    }
90 };

```

8.1.4 LiChaoTree

```

1 // To get an instance : node *root = EMPTY;
2 typedef pair<long long, long long> line;
3 int start_x, end_x;
4 line neutral(0, -1e18);
5
6 ll sub(const line &l1, ll x) {
7     return l1.first * x + l1.second;
8 }
9
10 double inter(const line &l1, const line &l2) {
11     return (l2.second - l1.second) / (l1.first - l2.first - .0);
12 }
13
14 extern struct node *const EMPTY;
15 struct node {
16     line l;
17     node *lf, *rt;
18     node() : l(neutral), lf(this), rt(this) {}
19     node(line l) : l(l), lf(EMPTY), rt(EMPTY) {}
20 };
21 node *const EMPTY = new node;
22

```

```

23 void insert(line l, int lx, int rx, node *&cur, ll ns = start_x, ll ne
  = end_x) {
24     if (rx < ns || lx > ne) return;
25     if (ns >= lx && ne <= rx) {
26         if (cur == EMPTY) {
27             cur = new node(l);
28             return;
29         }
30         if (l.first == cur->l.first) {
31             cur->l = max(cur->l, l);
32             return;
33         }
34
35         auto x = inter(l, cur->l);
36         if (x < ns || x > ne) {
37             if (sub(l, ns) > sub(cur->l, ns)) cur->l = l;
38             return;
39         }
40         ll mid = ns + (ne - ns) / 2;
41         if (x < mid) {
42             if (sub(l, ne) > sub(cur->l, ne)) swap(l, cur->l);
43             insert(l, lx, rx, cur->lf, ns, mid);
44         } else {
45             if (sub(l, ns) > sub(cur->l, ns)) swap(l, cur->l);
46             insert(l, lx, rx, cur->rt, mid + 1, ne);
47         }
48         return;
49     }
50
51     if (cur == EMPTY) {
52         cur = new node(neutral);
53     }
54     ll mid = ns + (ne - ns) / 2;
55     insert(l, lx, rx, cur->lf, ns, mid);
56     insert(l, lx, rx, cur->rt, mid + 1, ne);
57 }
58
59 ll query(ll x, node *cur, ll ns = start_x, ll ne = end_x) {
60     auto ret = sub(cur->l, x);
61     if (x < ns || x > ne)
62         return sub(neutral, x);
63     if (ns == ne)
64         return ret;
65     ll mid = ns + (ne - ns) / 2;
66     if (x <= mid)
67         return max(ret, query(x, cur->lf, ns, mid));
68     return max(ret, query(x, cur->rt, mid + 1, ne));
69 }

```

8.1.5 PersistentLiChaoTree

```

1 const ll maxN = 1e7 + 5;
2
3 struct Line {
4     ll m, c;
5
6     Line() : m(0), c(llinf) {}
7
8     Line(ll m, ll c) : m(m), c(c) {}
9 };
10
11 ll sub(ll x, Line l) {
12     return x * l.m + l.c;
13 }
14
15 // Persistent Li Chao
16 struct Node {
17     // range I am responsible for
18     Line line;
19     Node *left, *right;
20
21     Node() {
22         left = right = NULL;
23     }
24
25     Node(ll m, ll c) {
26         line = Line(m, c);
27         left = right = NULL;
28     }
29
30     void extend(int l, int r) {
31         if (left == NULL && l != r) {
32             left = new Node();
33             right = new Node();
34         }
35     }
36
37     Node *copy(Node *node) {
38         Node *newNode = new Node;
39         newNode->left = node->left;
40         newNode->right = node->right;
41         newNode->line = node->line;
42         return newNode;
43     }
44
45     Node *add(Line toAdd, int l, int r) {
46         int mid = (l + r) / 2;
47         Node *cur = copy(this);
48         if (l == r) {
49             if (sub(l, toAdd) < sub(l, cur->line))
50                 swap(toAdd, cur->line);
51             return cur;
52         }
53         bool lef = sub(l, toAdd) < sub(l, cur->line);
54         bool midE = sub(mid + 1, toAdd) < sub(mid + 1, cur->line);
55         if (midE)
56             swap(cur->line, toAdd);
57         cur->extend(l, r);
58         if (lef != midE) {
59             cur->left = cur->left->add(toAdd, l, mid);
60         } else {
61             if (mid + 1 > r)
62                 return cur;
63             cur->right = cur->right->add(toAdd, mid + 1, r);
64         }
65     }
66 }

```

```

64     }
65     return cur;
66 }
67
68 Node *add(Line toAdd) {
69     return add(toAdd, -maxN + 1, maxN - 1);
70 }
71
72 ll query(ll x, int l, int r) {
73     int mid = (l + r) / 2;
74     if (l == r || left == NULL)
75         return sub(x, line);
76     extend(l, r);
77     if (x <= mid)
78         return min(sub(x, line), left->query(x, l, mid));
79     else
80         return min(sub(x, line), right->query(x, mid + 1, r));
81 }
82
83 ll query(ll x) {
84     return query(x, -maxN + 1, maxN - 1);
85 }
86
87 void clear() {
88     if (left != NULL) {
89         left->clear();
90         right->clear();
91     }
92     delete this;
93 }
94 };
95
96 Node *tree[N];

```

8.2 Bitmask DP

8.2.1 SOS_DP

```

1 // Computes the sum of all subsets for every bitmask in O(N * 2^N)
2 vector<long long> SOS_DP(int n_bits, const vector<long long>& a) {
3     int max_mask = 1 << n_bits;
4     vector<long long> dp = a; // dp[mask] initially holds exactly the
5         value for mask
6
7     for (int i = 0; i < n_bits; ++i) {
8         for (int mask = 0; mask < max_mask; ++mask) {
9             if (mask & (1 << i)) {
10                 dp[mask] += dp[mask ^ (1 << i)];
11             }
12         }
13     }
14     return dp; // dp[mask] now contains sum of all submasks of mask

```

9 Trees

9.1 Lowest Common Ancestor (LCA)

9.1.1 LCA

```

1 struct LCA {
2     static const int B = 21;
3     vector<array<int, 2>> tour;
4     vector<array<array<int, 2>, B>> T;
5     vector<int> d, in, lg;
6
7     LCA(int n, auto &adj): d(n + 1), in(n + 1), lg(n*2 + 1) {
8         tour.reserve(n * 2);
9         T.resize(n);
10        dfs(0, 0, adj);
11        for(int i = 0; i < n; ++i) lg[i]=__lg(i), T[i][0]=tour[i];
12        for (int j = 1; j < B; ++j)
13            for (int i = 0; i + (1 << j) - 1 < n; ++i)
14                T[i][j] = min(T[i][j - 1], T[i+(1 << j) - 1][j - 1]);
15    }
16
17    void dfs(int u, int p, auto &adj) {
18        in[u] = size(tour);
19        tour.push_back({d[u], u});
20        for (int v: adj[u]) {
21            if (v == p) continue;
22            d[v] = d[u] + 1;
23            dfs(v, u, adj);
24            tour.push_back({d[u], u});
25        }
26    }
27
28    int get(int u, int v) {
29        int l = min(in[u], in[v]), r = max(in[u], in[v]);
30        int j = lg[r - l + 1];
31        return min(T[l][j], T[r - (1 << j) + 1][j])[1];
32    }
33
34    int dist(int u, int v) { return d[u]+d[v]-2*d[get(u, v)]; }
35 };

```

9.2 DSU on Tree

9.2.1 DSUOnTree

```

1 class DSUOnTree {
2 private:
3     struct Query {
4         int idx, x;
5     };
6
7     int n;
8     vector<vector<int>> adj;
9     vector<int> sz, depth;
10    vector<bool> big;
11    vector<vector<Query>> queries;
12    vector<int> ans;
13

```

```

14 // =====
15 // PROBLEM SPECIFIC STATE VARIABLES
16 // =====
17 vector<char> c; // Example: Array of characters per node
18
19 // Example state variables
20 // int distinct_count = 0;
21 // vector<int> freq;
22
23 void add(int u, int p) {
24     // --- ADD LOGIC HERE ---
25     // Example: freq[c[u] - 'a']++;
26
27     for (int v : adj[u]) {
28         if (v == p || big[v]) continue; // Skip parent and heavy
29         child[cite: 4]
30         add(v, u);
31     }
32 }
33
34 void remove(int u, int p) {
35     // --- REMOVE LOGIC HERE ---
36     // Example: freq[c[u] - 'a']--;
37
38     for (int v : adj[u]) {
39         if (v == p || big[v]) continue; // Skip parent and heavy
40         child[cite: 4]
41         remove(v, u);
42     }
43 }
44
45 int get_answer(int u, int x = 0) {
46     // --- QUERY ANSWER LOGIC HERE ---
47     return 0;
48 }
49 // =====
50
51 void pre(int u, int p = 0) {
52     sz[u] = 1;
53     depth[u] = depth[p] + 1;
54     for (int v : adj[u]) {
55         if (v == p) continue;
56         pre(v, u);
57         sz[u] += sz[v]; // Compute subtree size[cite: 4]
58     }
59 }
60
61 void dfs(int u, int p = 0, bool keep = false) {
62     int mx = -1, bigChild = -1;
63
64     // Find the heavy child[cite: 4]
65     for (int v : adj[u]) {
66         if (v == p) continue;
67         if (sz[v] > mx) {
68             mx = sz[v];
69             bigChild = v;
70         }
71     }
72
73     // Process light children and clear them[cite: 4]
74     for (int v : adj[u]) {
75         if (v == p || v == bigChild) continue;
76         dfs(v, u, false);
77     }
78
79     // Process heavy child and keep it[cite: 4]
80     if (bigChild != -1) {
81         dfs(bigChild, u, true);
82         big[bigChild] = true;
83     }
84
85     // Add current node and its light children[cite: 4]
86     add(u, p);
87
88     // Answer all queries for the current node[cite: 4]
89     for (const auto& q : queries[u]) {
90         ans[q.idx] = get_answer(u, q.x);
91     }
92
93     // Reset the big child flag[cite: 4]
94     if (bigChild != -1) {
95         big[bigChild] = false;
96     }
97
98     // Clear state if this node is not meant to be kept[cite: 4]
99     if (!keep) {
100        remove(u, p);
101    }
102 }
103
104 public:
105 // Initialize with 1-based indexing in mind
106 DSUOnTree(int nodes, const vector<char>& chars) : n(nodes), c(chars) {
107     adj.assign(n + 1, vector<int>());
108     sz.assign(n + 1, 0);
109     depth.assign(n + 1, 0);
110     big.assign(n + 1, false);
111     queries.assign(n + 1, vector<Query>());
112     // Initialize problem-specific arrays here (e.g., freq.assign
113         (26, 0);)
114 }
115
116 void add_edge(int u, int v) {
117     adj[u].push_back(v);
118     adj[v].push_back(u);
119 }
120
121 void add_query(int node, int query_idx, int x = 0) {
122     queries[node].push_back({query_idx, x});
123 }
124
125 vector<int> solve(int root, int num_queries) {
126     ans.assign(num_queries, 0);

```

```
124     pre(root, 0); // Precalculate sizes and depths[cite: 4]
125     dfs(root, 0, false); // Run the sack algorithm[cite: 4]
126     return ans;
127 }
128 };
```

9.3 Heavy Light Decomposition (HLD)

9.3.1 HLD

```
1 class HLD {
2 public:
3     vector<int> par, sz, head, tin, tout, who, depth;
4
5     int dfs1(int u, vector<vector<int>> &adj) {
6         for (int &v: adj[u]) {
7             if (v == par[u]) continue;
8             depth[v] = depth[u] + 1;
9             par[v] = u;
10            sz[u] += dfs1(v, adj);
11            if (sz[v] > sz[adj[u][0]] || adj[u][0] == par[u]) swap(v,
12                adj[u][0]);
13        }
14        return sz[u];
15    }
16
17    void dfs2(int u, int &timer, const vector<vector<int>> &adj) {
18        tin[u] = timer++;
19        for (int v: adj[u]) {
20            if (v == par[u]) continue;
21            head[v] = (timer == tin[u] + 1 ? head[u] : v);
22            dfs2(v, timer, adj);
23        }
24        tout[u] = timer - 1;
25    }
26
27    HLD(vector<vector<int>> &adj, int r = 0)
28        : par(adj.size(), -1), sz(adj.size(), 1), head(adj.size(), r),
29          tin(adj.size(), 0), tout(adj.size(), 0),
30          who(adj.size(), 0), depth(adj.size(), 0) {
31        dfs1(r, adj);
32        int x = 0;
33        dfs2(r, x, adj);
34        for (int i = 0; i < adj.size(); ++i) who[tin[i]] = i;
35    }
36
37    vector<pair<int, int>> path(int u, int v) {
38        vector<pair<int, int>> res;
39        for (; v = par[head[v]] {
40            if (depth[head[u]] > depth[head[v]]) swap(u, v);
41            if (head[u] != head[v]) {
42                res.emplace_back(tin[head[v]], tin[v]);
43            } else {
44                if (depth[u] > depth[v]) swap(u, v);
45                res.emplace_back(tin[u], tin[v]);
46                return res;
47            }
48        }
49
50        pair<int, int> subtree(int u) {
51            return {tin[u], tout[u]};
52        }
53
54        int dist(int u, int v) {
55            return depth[u] + depth[v] - 2 * depth[lca(u, v)];
56        }
57
58        int lca(int u, int v) {
59            for (; v = par[head[v]] {
60                if (depth[head[u]] > depth[head[v]]) swap(u, v);
61                if (head[u] == head[v]) {
62                    if (depth[u] > depth[v]) swap(u, v);
63                    return u;
64                }
65            }
66        }
67
68        bool isAncestor(int u, int v) {
69            return tin[u] <= tin[v] && tout[u] >= tout[v];
70        }
71 };
```

9.4 Centroid Decomposition

9.4.1 Centroid

```
1 class CentroidDecomposition {
2 private:
3     int n;
4     vector<vector<int>> adj;
5     vector<int> sz, par, nodes;
6     vector<bool> vis;
7
8     int dfsSz(int u, int p) {
9         sz[u] = 1;
10        for (int v: adj[u]) {
11            if (vis[v] || v == p) continue;
12            sz[u] += dfsSz(v, u);
13        }
14        return sz[u];
15    }
16
17    int getCentroid(int u, int p, int total_nodes) {
18        for (int v: adj[u]) {
19            if (vis[v] || v == p) continue;
20            if (sz[v] * 2 > total_nodes) {
21                return getCentroid(v, u, total_nodes);
22            }
23        }
24        return u;
25    }
26 }
```

```
27 void collect(int u, int p) {
28     nodes.push_back(u);
29     for (int v: adj[u]) {
30         if (vis[v] || v == p) continue;
31         collect(v, u);
32     }
33 }
34
35 void build(int u, int p) {
36     int total_nodes = dfsSz(u, p);
37     int centroid = getCentroid(u, p, total_nodes);
38
39     if (p != -1) {
40         par[centroid] = p;
41     } else {
42         par[centroid] = centroid;
43     }
44
45     vis[centroid] = true;
46
47     // =====
48     // PROBLEM SPECIFIC PROCESSING
49     // =====
50     // Process paths going through the 'centroid' here
51     for (int v: adj[centroid]) {
52         if (vis[v]) continue;
53         nodes.clear();
54         collect(v, centroid);
55
56         // e.g., Update answers using the nodes collected from this
57         // subtree branch
58     }
59     // =====
60
61     // Recursively decompose the remaining components
62     for (int v: adj[centroid]) {
63         if (vis[v]) continue;
64         build(v, centroid);
65     }
66 }
67 public:
68     // Initialize with 1-based indexing
69     CentroidDecomposition(int nodes_count) : n(nodes_count) {
70         adj.assign(n + 1, vector<int>());
71         sz.assign(n + 1, 0);
72         par.assign(n + 1, 0);
73         vis.assign(n + 1, false);
74     }
75
76     void add_edge(int u, int v) {
77         adj[u].push_back(v);
78         adj[v].push_back(u);
79     }
80
81     // Start decomposition, usually from node 1
82     void decompose(int root = 1) {
83         build(root, -1);
84     }
85
86     // Returns the parent of a node in the centroid tree
87     int get_centroid_parent(int u) const {
88         return par[u];
89     }
90 };
```

10 Game Theory

11 Geometry

11.1 Geometry 2D

11.1.1 Point2D

```
1 // 2D Point / Vector Primitive
2 template <typename T = double>
3 struct Point {
4     typedef Point P;
5     T x, y;
6
7     Point(T x = 0, T y = 0) : x(x), y(y) {}
8
9     bool operator<(P p) const { return tie(x, y) < tie(p.x, p.y); }
10
11     bool operator==(P p) const { return tie(x, y) == tie(p.x, p.y); }
12
13     P operator+(P p) const { return P(x + p.x, y + p.y); }
14
15     P operator-(P p) const { return P(x - p.x, y - p.y); }
16
17     P operator*(T d) const { return P(x * d, y * d); }
18
19     P operator/(T d) const { return P(x / d, y / d); }
20
21     T dot(P p) const { return x * p.x + y * p.y; }
22
23     T cross(P p) const { return x * p.y - y * p.x; }
24
25     T cross(P a, P b) const { return (a - *this).cross(b - *this); }
26
27     T dist2() const { return x * x + y * y; }
28
29     T dist2(P b) const { return (*this - b).dist2(); }
30
31     double dist() const { return sqrt((double) dist2()); }
32
33     double dist(P b) { return (*this - b).dist(); }
34
35     // angle to x-axis in interval [-pi, pi]
36     double angle() const { return atan2(y, x); }
37
38     P unit() const { return *this / dist(); } // makes dist()=1
39     P perp() const { return P(-y, x); } // rotates +90 degrees
40     P normal() const { return perp().unit(); }
```

```

41 P getVector(const P &p) const { return p - (*this); }
42
43 // returns point rotated 'a' radians ccw around the origin
44 P rotate(double a) const {
45     return P(x * cos(a) - y * sin(a), x * sin(a) + y * cos(a));
46 }
47
48 friend ostream &operator<<(ostream &os, P p) {
49     return os << "(" << p.x << ", " << p.y << ")";
50 }
51 //
52 return os << p.x << " " << p.y;
53 }
54
55 friend istream &operator>>(istream &is, P &p) {
56     return is >> p.x >> p.y;
57 }
58
59 // Project point onto line through a and b (assuming a != b).
60 P projectOnLine(const P &a, const P &b) const {
61     P ab = a.getVector(b);
62     P ac = a.getVector(*this);
63     return a + ab * ac.dot(ab) / a.dist2(b);
64 }
65
66 // Project point c onto line segment through a and b (assuming a !=
67 // b).
68 P projectOnSegment(const P &a, const P &b) const {
69     P ab = b - a, ac = *this - a;
70     long double r = ac.dot(ab), d = a.dist2(b);
71     if (r < 0) return a;
72     if (r > d) return b;
73     return a + ab * r / d;
74 }
75
76 P reflectAroundLine(const P &a, const P &b) const {
77     return projectOnLine(a, b) * 2 - *this;
78 };
79
80 int dcmp(const ld &a, const ld &b){
81     if(fabs(a - b) < eps)
82         return 0;
83
84     return (a > b ? 1 : -1);
85 }

```

11.1.2 Line2D

```

1 template<class T>
2 double lineDist(T p, T s, T e) {
3     if (s == e) {
4         return s.dist(p);
5     }
6     return fabs((p - s).cross(e - s) / (e - s).dist());
7 }
8
9 template<class T>
10 pair<int, T> lineInter(T s1, T e1, T s2, T e2) {
11     // first = 0 no intersection
12     // first = 1 intersection
13     // first = -1 infinite intersection
14     auto d = (e1 - s1).cross(e2 - s2);
15     if (dcmp(d,0) == 0) // if parallel same line first = -1 else
16         first = 0
17         return {-(s1.cross(e1, s2) == 0), {0,0}};
18     auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
19     return {1, (s1 * p + e1 * q) / d};
20 }
21
22 template<class T>
23 bool onSegment(T p,T s, T e) {
24     return dcmp(p.cross(s, e),0) == 0 && dcmp((s - p).dot(e - p), 0)
25         <= 0;
26 }
27
28 template<class T>
29 double segDist(T p, T s, T e) {
30     if (dcmp((p - s).dot(e - s),0) <= 0) return s.dist(p);
31     if (dcmp((p - e).dot(e - s),0) >= 0) return e.dist(p);
32     return lineDist(p, s, e);
33 }
34
35 template<class T>
36 pair<int, T> segInter(T s1, T e1, T s2, T e2) {
37     // first = 0 no intersection
38     // first = 1 intersection
39     // first = -1 infinite intersection
40     pair<int, T> ret = lineInter(s1, e1, s2, e2);
41     if (ret.first == 0) return ret;
42     else if (ret.first == 1) {
43         if (onSegment(ret.second, s1, e1) && onSegment(ret.second, s2,
44             e2))
45             return ret;
46         else
47             return {0, {0, 0}};
48     } else {
49         if (onSegment(s1, s2, e2) || onSegment(e1, s2, e2))
50             return {-1, (onSegment(s1, s2, e2) ? s1 : e1)};
51         else if (onSegment(s2, s1, e1) || onSegment(e2, s1, e1))
52             return {-1, (onSegment(s2, s1, e1) ? s2 : e2)};
53         else
54             return {0, {0, 0}};
55     }
56 }
57
58 template<class T>
59 T closestOnSegment(T p, T s, T e) {
60     if ((p - s).dot(e - s) <= 0) return s;
61     else if ((p - e).dot(e - s) >= 0) return e;
62

```

```

63     else return p.projectOnLine(s, e);
64 }
65
66 template<class T>
67 double segSegDist(T s1, T e1, T s2, T e2) {
68     if(segInter(s1,e1,s2,e2).first != 0)
69         return 0;
70     double ret = min({segDist(s1,s2,e2), segDist(e1,s2,e2), segDist(s2,
71         s1,e1), segDist(e2,s1,e1)});
72     return ret;
73 }
74
75 template<class T>
76 bool onRay(T p, T s, T e) {
77     return dcmp(p.cross(s, e), 0) == 0 && dcmp((p - s).dot(e - s), 0)
78         >= 0;
79 }
80
81 template<class T>
82 double rayDist(T p, T s, T e) {
83     if ((p - s).dot(e - s) <= 0) {
84         return s.dist(p);
85     }
86     return lineDist(p, s, e);
87 }
88
89 template<class T>
90 pair<int, T> rayInter(T s1, T e1, T s2, T e2) {
91     // first = 0 no intersection
92     // first = 1 intersection
93     // first = -1 infinite intersection
94     pair<int, T> ret = lineInter(s1, e1, s2, e2);
95     if (ret.first == 0) return ret;
96     if (ret.first == 1) {
97         if (onRay(ret.second, s1, e1) && onRay(ret.second, s2, e2))
98             return ret;
99         else
100             return {0, {0,0}};
101     } else {
102         if (onRay(s1, s2, e2) || onRay(s2, s1, e1))
103             return {-1, onRay(s1, s2, e2) ? s1 : s2};
104         else
105             return {0, {0,0}};
106     }
107 }
108
109 template<class T>
110 double rayRayDist(T s1, T e1, T s2, T e2) {
111     if(rayInter(s1,e1,s2,e2).first != 0)
112         return 0;
113     double ret = min(rayDist(s1,s2,e2), rayDist(s2,s1,e1));
114     return ret;
115 }
116

```

11.1.3 AngleOperations

```

1 // Angle Operations in 2D - [-pi, pi] and [0, 2*pi]
2 template<typename T>
3 ld angleBetween(T a, T b) {
4     ld res = atan2(a.cross(b), a.dot(b));
5     return res < 0 ? res + 2 * PI : res;
6 }
7
8 template<typename T>
9 ld angle0(T a, T 0, T b) {
10     return angleBetween(a - 0, b - 0);
11 }
12
13 ld formatAngle(ld angle) {
14     while (angle > PI) angle -= 2 * PI;
15     while (angle <= -PI) angle += 2 * PI;
16     return angle;
17 }

```

11.1.4 PolygonArea

```

1 // Polygon Area (Shoelace Formula) - O(N)
2 template<typename T>
3 ld polygonArea(const vector<T>& v) {
4     ld area = 0;
5     int n = v.size();
6     for (int i = 0; i < n; i++) {
7         area += v[i].cross(v[(i + 1) % n]);
8     }
9     return 0.5 * abs(area);
10 }

```

11.1.5 Circle3Points

```

1 // Circumcenter & Circumradius of 3 Points
2 typedef Point<double> P;
3
4 bool isColliner(const P &A, const P &B, const P &C) {
5     return dcmp(P(B - A).cross(P(C - A)), 0) == 0;
6 }
7
8 P ccCenter(const P &A, const P &B, const P &C) {
9     P b = C - A, c = B - A;
10     return A + (b * c.dist2() - c * b.dist2()).perp() / b.cross(c) / 2;
11 }
12
13 double ccRadius(const P &A, const P &B, const P &C) {
14     return (B - A).dist() * (C - B).dist() * (A - C).dist() / (2 * abs
15         ((B - A).cross(C - A)));
16 }

```

11.1.6 CircleLineIntersection

```

1 template<class T>
2 vector<T> circleLine(T o, double r, T a, T b) {

```

```

3     double h2 = r * r - lineDist(o, a, b) * lineDist(o, a, b);
4     if (dcmp(h2, 0) != -1) { // the line touches the circle
5         T p = o.projectOnLine(a, b); // point P
6         T h = (a - b).unit() * sqrt(h2); // vector parallel to l, of
7             length h
8         if (p - h == p + h)
9             return {p - h};
10        else
11            return {p - h, p + h};
12    }
13    return {};
14 }
15 template<class T>
16 vector<T> circleSegment(T o, double r, T a, T b) {
17     vector<T> v = circleLine(o, r, a, b);
18     vector<T> out;
19     for (auto x: v)
20         if (onSegment(x, a, b))
21             out.pb(x);
22     return out;
23 }

```

11.1.7 CircleCircleIntersection

```

1 template<class T>
2 ld circleInter(T a, T b, ll r1, ll r2) {
3     if (r1 > r2) {
4         swap(r1, r2);
5         swap(a, b);
6     }
7     ld d = sqrt((a.x - b.x) * (a.x - b.x) + (a.y - b.y) * (a.y - b.y));
8     if (d >= r1 + r2) {
9         return 0;
10    }
11    if (d + r1 <= r2) {
12        return PI * r1 * r1;
13    }
14    ld theta = acos((d * d + r2*r2 - r1*r1)/(2 * d * r2));
15    ld alpha = acos((d * d + r1*r1 - r2*r2)/(2 * d * r1));
16    ld s1 = theta * r2 * r2;
17    ld t1 = 0.5 * r2 * r2 * sin(2 * theta);
18    ld s2 = alpha * r1 * r1;
19    ld t2 = 0.5 * r1 * r1 * sin(2 * alpha);
20    return (s1 - t1) + (s2 - t2);
21 }

```

11.1.8 ConvexHull

```

1 // Convex Hull (Andrew's Monotone Chain) - O(N log N)
2 // Returns vertices in counter-clockwise order
3 template<typename P>
4 vector<P> convexHull(vector<P> pts) {
5     int n = pts.size(), k = 0;
6     if (n <= 2) return pts;
7     vector<P> h(2 * n);
8     sort(all(pts));
9     for (int i = 0; i < n; i++) {
10        while (k >= 2 && h[k - 2].cross(h[k - 1], pts[i]) <= 0) k--;
11        h[k++] = pts[i];
12    }
13    for (int i = n - 2, t = k + 1; i >= 0; i--) {
14        while (k >= t && h[k - 2].cross(h[k - 1], pts[i]) <= 0) k--;
15        h[k++] = pts[i];
16    }
17    h.resize(k - 1);
18    return h;
19 }

```

11.1.9 RotatingCalipers

```

1 // Rotating Calipers - O(N) Convex Polygon Diameter & Width
2 template<typename T>
3 ll hullDiameter(const vector<T>& S) {
4     int n = S.size(), j = n < 2 ? 0 : 1;
5     ll res = 0;
6     for (int i = 0; i < j; i++) {
7         for (; j = (j + 1) % n; ) {
8             res = max(res, (ll)(S[i] - S[j]).dist2());
9             if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
10                break;
11        }
12    }
13    return res;
14 }
15 template<typename T>
16 ld hullWidth(const vector<T>& S) {
17     int n = S.size();
18     if (n <= 2) return 0;
19     int j = 1;
20     ld res = 1e18;
21     for (int i = 0; i < n; i++) {
22         while ((S[(i + 1) % n] - S[i]).cross(S[(j + 1) % n] - S[j]) >=
23             0) j = (j + 1) % n;
24         res = min(res, lineDist(S[j], S[i], S[(i + 1) % n]));
25     }
26     return res;
27 }

```

11.1.10 MinimumEnclosingCircle

```

1 // Minimum Enclosing Circle (Welzl's Algorithm) - O(N) Expected Time
2 typedef Point<double> P;
3
4 P ccCenter(const P& A, const P& B, const P& C) {
5     P b = C - A, c = B - A;
6     return A + (b * c.dist2() - c * b.dist2()).perp() / b.cross(c) / 2;
7 }
8
9 pair<P, double> mec(vector<P> ps) {
10    shuffle(all(ps), mt19937(chrono::steady_clock::now().
11        time_since_epoch().count()));

```

```

11 P o = ps[0];
12 double r = 0, EPS = 1 + 1e-9;
13 int n = ps.size();
14 for (int i = 0; i < n; i++) {
15     if ((o - ps[i]).dist() > r * EPS) {
16         o = ps[i]; r = 0;
17         for (int j = 0; j < i; j++) {
18             if ((o - ps[j]).dist() > r * EPS) {
19                 o = (ps[i] + ps[j]) / 2;
20                 r = (o - ps[i]).dist();
21                 for (int k = 0; k < j; k++) {
22                     if ((o - ps[k]).dist() > r * EPS) {
23                         o = ccCenter(ps[i], ps[j], ps[k]);
24                         r = (o - ps[i]).dist();
25                     }
26                 }
27             }
28         }
29     }
30 }
31 return {o, r};
32 }

```

11.1.11 HalfPlaneIntersection

```

1 // Basic half-plane struct.
2 template<class T>
3 struct Halfplane {
4
5     // 'p' is a passing point of the line and 'pq' is the direction
6     vector of the line.
7     T p, pq;
8     long double angle;
9
10    Halfplane() {}
11    Halfplane(const T& a, const T& b) : p(a), pq(b - a) {
12        angle = atan2l(pq.y, pq.x);
13    }
14
15    // Check if point 'r' is outside this half-plane.
16    // Every half-plane allows the region to the LEFT of its line.
17    bool out(const T& r) {
18        return dcmp(pq.cross(r - p), 0) < 0;
19    }
20
21    // Comparator for sorting.
22    bool operator < (const Halfplane& e) const {
23        return angle < e.angle;
24    }
25
26    // Intersection point of the lines of two half-planes. It is
27    assumed they're never parallel.
28    friend T inter(const Halfplane& s, const Halfplane& t) {
29        long double alpha = (ld)(t.p - s.p).cross(t.pq) / s.pq.cross(t.pq);
30        return s.p + (s.pq * alpha);
31    }
32 };
33 template<class T>
34 vector<T> hpIntersect(vector<Halfplane<T>> H) {
35     ld inf = 1e9;
36     T box[4] = { // Bounding box in CCW order
37         Point(inf, inf),
38         Point(-inf, inf),
39         Point(-inf, -inf),
40         Point(inf, -inf)
41     };
42
43     for (int i = 0; i < 4; i++) { // Add bounding box half-planes.
44         Halfplane aux(box[i], box[(i + 1) % 4]);
45         H.push_back(aux);
46     }
47
48     // Sort by angle and start algorithm
49     sort(H.begin(), H.end());
50     deque<Halfplane<T>> dq;
51     int len = 0;
52     for (int i = 0; i < int(H.size()); i++) {
53
54         // Remove from the back of the deque while last half-plane is
55         redundant
56         while (len > 1 && H[i].out(inter(dq[len - 1], dq[len - 2]))) {
57             dq.pop_back();
58             --len;
59         }
60
61         // Remove from the front of the deque while first half-plane is
62         redundant
63         while (len > 1 && H[i].out(inter(dq[0], dq[1]))) {
64             dq.pop_front();
65             --len;
66         }
67
68         // Special case check: Parallel half-planes
69         if (len > 0 && fabs1(H[i].pq.cross(dq[len - 1].pq)) < eps) {
70             // Opposite parallel half-planes that ended up checked
71             against each other.
72             if (dcmp(H[i].pq.dot(dq[len - 1].pq), 0) < 0)
73                 return vector<T>();
74
75             // Same direction half-plane: keep only the leftmost half-
76             plane.
77             if (H[i].out(dq[len - 1].p)) {
78                 dq.pop_back();
79                 --len;
80             }
81         }
82
83         // Add new half-plane
84         dq.push_back(H[i]);
85         ++len;
86     }

```

```

83 // Final cleanup: Check half-planes at the front against the back
84 // and vice-versa
85 while (len > 2 && dq[0].out(inter(dq[len-1], dq[len-2]))) {
86     dq.pop_back();
87     --len;
88 }
89
90 while (len > 2 && dq[len-1].out(inter(dq[0], dq[1]))) {
91     dq.pop_front();
92     --len;
93 }
94
95 // Report empty intersection if necessary
96 if (len < 3) return vector<T>();
97
98 // Reconstruct the convex polygon from the remaining half-planes.
99 vector<T> ret(len);
100 for(int i = 0; i < len; i++) {
101     ret[i] = inter(dq[i], dq[(i+1)%len]);
102 }
103 return ret;
104 }

```

11.2 Geometry 3D

11.2.1 Point3D

```

1 // 3D Point / Vector Primitive
2 template <typename T = double>
3 struct Point3D {
4     typedef Point3D P;
5     typedef const P &R;
6     T x, y, z;
7
8     explicit Point3D(T x = 0, T y = 0, T z = 0) : x(x), y(y), z(z) {}
9
10    bool operator<(R p) const {
11        return tie(x, y, z) < tie(p.x, p.y, p.z);
12    }
13
14    bool operator==(R p) const {
15        return tie(x, y, z) == tie(p.x, p.y, p.z);
16    }
17
18    P operator+(R p) const { return P(x + p.x, y + p.y, z + p.z); }
19
20    P operator-() const { return P(-x, -y, -z); }
21
22    P operator-(R p) const { return P(x - p.x, y - p.y, z - p.z); }
23
24    P operator*(T d) const { return P(x * d, y * d, z * d); }
25
26    P operator/(T d) const { return P(x / d, y / d, z / d); }
27
28    T dot(R p) const { return x * p.x + y * p.y + z * p.z; }
29
30    P cross(R p) const {
31        return P(y * p.z - z * p.y, z * p.x - x * p.z, x * p.y - y * p.x);
32    }
33
34    T dist2() const { return x * x + y * y + z * z; }
35
36    double dist() const { return sqrt((double) dist2()); }
37
38    //Azimuthal angle ( longitude ) to x=axis in interval [pi , pi ]
39    double phi() const { return atan2(y, x); }
40
41    //Zenith angle ( latitude ) to the z=axis in interval [0 , pi ]
42    double theta() const { return atan2(sqrt(x * x + y * y), z); }
43
44    P unit() const { return *this / (T) dist(); } //makes dist ()=1
45    //returns unit vector normal to *this and p
46    P normal(P p) const { return cross(p).unit(); }
47
48    //returns point rotated 'angle ' radians ccw around axis
49    P rotate(double angle, P axis) const {
50        double s = sin(angle), c = cos(angle);
51        P u = axis.unit();
52        return u * dot(u) * (1 - c) + (*this) * c - cross(u) * s;
53    }
54
55    friend ostream &operator<<(ostream &os, P p) {
56        // return os << "(" << p.x << ", " << p.y << ")";
57        return os << p.x << " " << p.y << " " << p.z;
58    }
59
60    friend istream &operator>>(istream &is, P &p) {
61        return is >> p.x >> p.y >> p.z;
62    }
63 };
64
65 namespace Axis {
66     Point3D<double> x(1, 0, 0);
67     Point3D<double> y(0, 1, 0);
68     Point3D<double> z(0, 0, 1);
69 }
70

```